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Review Article



Emerging technologies and supporting tools for earthquake disaster management: A perspective, challenges, and future directions

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ABSTRACT

Seismology is among the ancient sciences that concentrate on earthquake disaster management (EQDM), which directly impact human life and infrastructure resilience. Such a pivot has made use of contemporary technologies. Nevertheless, there is a need for more reliable and insightful solutions to tackle the daily challenges and intricacies of the natural sciences that stakeholders must confront. To consolidate the substantial endeavors in this field, we undertake a comprehensive survey of the interconnected contemporary technologies. More particularly, we analyze the data communication networks (DCNs) and the Internet of Things (IoT), which are among the main infrastructures of seismic networks. In accordance, we present conventional and innovative signal-processing techniques in seismology. Then, we shed light on the evolution of EQ sensors including the acoustic sensors based on optical fibers. Furthermore, we address the role of remote sensing (RS), robots, and drones for EQDM. Afterward, we highlight the social media contribution. Subsequently, a comprehensive elucidation of the diverse optimization techniques employed in seismology and for prolonging seismic networks is presented. Besides, the paper analyzes the important functions that artificial intelligence (AI) can fulfill in several areas of seismology. Lastly, we guide stakeholders on how to prevent natural disasters and preserve human lives.

1. Introduction

Understanding the intricate make-up and structure of our planet is crucial because of how complex it is. The Earth is composed of different strata that are interrelated and have different characteristics. Earthquakes (EQs) and seismic activity are caused by the non-homogeneity of the Earth's crust, which allows seismic waves to propagate across different layers. Enthusiasts in the field of seismology investigate the intricate phenomena of EQ waves and how they behave inside the Earth [1,2,3].

A variety of natural and man-made phenomena can cause vibrations that travel through the Earth's crust, a phenomenon known as a seismic wave. Scientists investigate the earth's innards with the help of these waves because they contain important information about the underlying structure. Important information can be retrieved from these waves with the use of seismic signal processing methods. For the purpose of improving and analyzing seismic data, these approaches use a number of mathematical and computational tools. Filtering, deconvolution, and migration are common processing techniques that help to improve resolution, detect subsurface features properly, and remove undesired frequencies and noise [4,5,6,7,8]. Reconstructing intricate pictures of underground structures also makes use of cutting-edge techniques like waveform inversion and tomography. Geoscientists are able to better understand seismic data, spot any dangers, and investigate natural resources with the use of seismic signal processing. Seismic exploration, EQ monitoring, and infrastructure integrity assessments in seismically active areas rely heavily on it. In addition, different types of seismic waves can be defined according to how they travel. The EQ waves are composed of several types. Among them, P-wave, S-wave, and surface wave are the most known ones. First, the P-wave can move through

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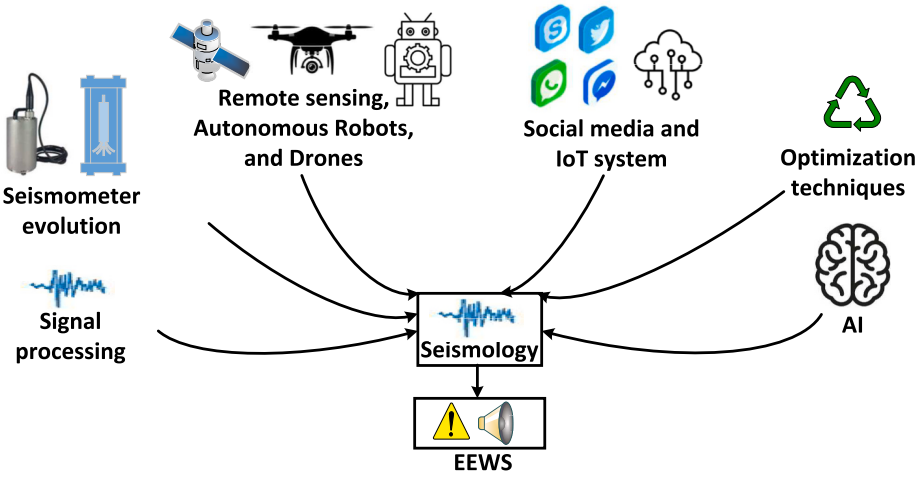


Fig. 1. Revolutionizing Seismology: Cutting-Edge Technologies and Innovations.

solids, liquids, and gases. Second and third, S- and surface waves, can move through only in solids. Indeed, the surface wave is the most destructive one among them, which can travel along the earth [9].

The seismic signal processing is to carefully analyze the EQ waves via different algorithms. This operation is also used to denoise the seismic data and check the data quality. Following this operation, filtering mechanisms are used to clarify the wanted signals and remove the unwanted frequency components [10]. It is worth mentioning that deconvolution is among the main techniques employed for seismic wave processing. This method can operate along with other techniques such as Wiener, least-squares, and wavelet. The deconvolution can be utilized for enhancing the signal resolution and clarify the hidden geological characteristics. This method is capable of filtering the effect of the seismic source and the subsurface of the earth from the wanted signals [11]. Interestingly, waveform inversion and tomography are among the modernized techniques that can be used for reconstructing the subsurface features. On the one hand, the waveform inversion model works iteratively till reaching the best parameter where the real observed data is compared with the calculated synthetic one in each iteration. On the other hand, the tomography technique operates by creating 3D pictures of subsurface structures based on the weak motion waves of the EQ. It contributes to figuring out both the complicated geological formations and possible fluid reservoirs [12].

EQ wave processing is not limited to geotechnical studies but is also important for minerals exploration, EQ prediction, urban expansion, and more. Scientists can progress in geology and engineering by using these methods, which will allow them to better understand the Earth's dynamic processes [13].

A multidisciplinary strategy integrating expertise in areas like mathematics, computer science, physics, electrical engineering, and others is necessary for comprehending and tracking EQs. Due to their multifaceted nature, EQs demand a wide range of methods and resources for a complete understanding of their inner workings. Thus, crucial technologies for seismic network support and EQ study facilitation have arisen, including the Internet of Things (IoT), data communication networks (DCNs), social networks, remote sensing (RS), drones, robots, optimization approaches, and artificial intelligence (AI) [14,15,16,17]. These networks, based on modern systems, play a pivotal role in the observation, transmission, and interconnection of seismic signals. These networks bridge the gap between the engineering aspects of EQ monitoring and the seismological aspects of understanding seismic waves. By employing these networks, researchers and scientists can gather valuable data about seismic networks and waves, leading to a deeper understanding of EQ phenomena. More particularly, IoT has expanded its applications beyond seismic studies, finding utility in earthquake early warning systems (EEWS), DM, and the mitigation of seismic hazards in regions prone to [18]. One of the main benefits is the ability to provide fast warning of strong shaking. By detecting and analyzing seismic signals in real-time, IoT can issue timely alerts, giving people in affected areas crucial seconds or even minutes to take protective measures. This early warning capability can significantly reduce the potential impact on infrastructure, human lives, and the overall well-being of communities. Furthermore, IoT exhibits swift processing times, enabling rapid data analysis and decision-making in response to seismic events. This efficiency is particularly valuable in situations where immediate actions are required to mitigate the effects. Another advantage of IoT is its effectiveness in blind zones, which are regions located close to the seismic source. In such areas, traditional detection techniques may not provide sufficient warning time due to the short distance between the source and the target site. However, IoT can overcome this limitation by leveraging their network infrastructure and advanced algorithms to quickly detect and transmit seismic signals, providing valuable warning time even in these challenging scenarios [19].

Moreover, the utilization of modern technologies extends beyond the immediate transmission of warning messages [20,21,22,23,24]. These networks can effectively leverage existing road infrastructures to manage disasters and mitigate the resulting crises. Merging modern technologies with the transportation system can assist the decision-maker in taking suitable actions for road network crises and during an EQ disaster. Such technologies can support the administration in facilitating an effective evacuation plan for the disastrous area by supporting the desired real-time statistics and status. Modern technologies have shortened the time needed to get the observed seismic data. More particularly, these technologies enhanced the EWS and data transfer via emergency protocols [25]. There are many tools employed to enhance understanding of the complex structure of the earth, EQ disaster, prolonging the lifetime of EQ network, and improving the seismic data transmission. Among these tools, optimization plays a significant role not only in the above-mentioned aspects but also in seismic hazard assessment, seismic data interpretation, etc. Besides, optimization techniques have been employed for determining the optimal packet transmission [25,26,27,28,29,30]. Moreover, these techniques have been involved in securing data privacy and authenticity [31,32,33,34,35,36], leading to faster and more reliable data interchange [26,37,38,39]. Recently, AI has evolved to discriminate between different seismic events and alleviate the EQ disaster, which can be easily migrated with different technologies. In order to detect patterns, outliers, and precursors to seismic occurrences, AI algorithms can sift through mountains of seismic data. By leveraging machine learning (ML) and pattern recognition algorithms, AI can contribute to the early detection and accurate characterization of seismic events. This information can assist in predicting the intensity and potential damage of

Table 1
Main abbreviations.

Original Term	Abbreviation	Original Term	Abbreviation
Affinity propagation	AP	Artificial intelligence	AI
Deep autoencoder	DAE	Akaike information criterion	AIC
Data communication networks	DCN	Carrier sense multiple access with collision detection	CSMA/CD
Deep learning	DL	Deep convolutional neural network	DCNN
Distributed acoustic sensing	DAS	Disaster recovery system	DRS
Distance vector	DV	Deep neural network	DNN
Earthquake early warning system	EEWS	Density-based spatial clustering of applications with noise	HDBSCAN
Long short-term memory	LSTM	Earthquake	EQ
Machine learning	ML	Internet of things	IoT
Peak ground acceleration	PGA	Long-term average	LTA
Particle swarm optimization	PSO	Peak particle velocity	PPV
Software-defined network	SDN	Quarry blast	QB
Support vector machine	SVM	Seismic activities monitoring and analysis system	SAMS
Time division multiple access	TDMA	Sensor nodes	SNs
Supervised and Unsupervised	S and US	Short-term average	STA

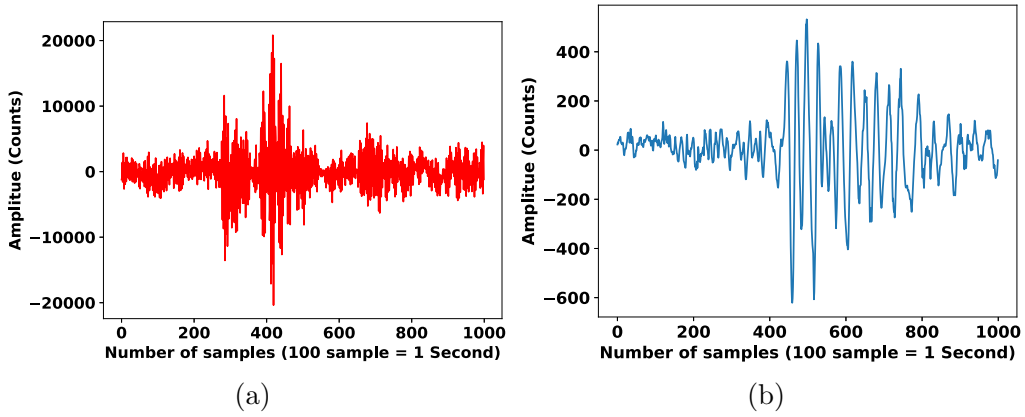


Fig. 2. (a) EQ wave, (b) QB wave.

affected areas by EQs, enabling authorities to issue timely warnings and implement appropriate disaster preparedness measures [37,26].

Although different research efforts and surveys have been presented in the literature, they lack the integrity of the emerging technologies for the EQ. DM. Moreover, to the best of our knowledge, there is no similar inclusive study with deep insights that has been discussed before. This paper provides a comprehensive investigation of the interdependence between EQ. DM and emerging technologies, including DCNs, signal processing, EQ sensors, RS, autonomous robots, drones, social networks, optical fiber, IoT, optimization techniques, and AI. The authors discuss these technologies and tools from various aspects, emphasizing the importance of considering the specific characteristics of the region and the desired objectives when selecting an approach [40,41,42]. These emerging technologies can contribute either separately or in conjunction to attain the most optimal outcome for EQ. DM [43,44,45].

The structure of the paper is organized as follows: Section 2 offers an overview of the importance of modern technologies in seismology. Section 3 gives an overview of seismic waves and seismic signal processing techniques. Section 5 highlights the role of RS in seismic studies. Section 6 discusses the roles of social networks and IoT for various seismology applications. Section 7 provides an in-depth review of different optimization techniques employed in seismic wave analysis and network longevity. Section 8 summarizes the significant contributions of AI in various disciplines within seismology. In Section 9, we present six case studies that utilize different real-world seismic datasets. Finally, in Section 10, the paper concludes with recommendations for stakeholders in the field. The paper's flow is depicted in Fig. 1. (See Table 1 that provide a list of abbreviations.)

2. Seismology the digital age: embracing the synergy of emerging technologies

The EQ or QB has the potential to generate an elastic wave called the seismic wave as shown in Fig. 2. These waves can be categorized into different types, such as Rayleigh and Love waves, which propagate through both the Earth's interior and its surface as P and S waves, respectively. They carry valuable information about the Earth's hidden features, including monument areas, nonrenewable energy sources, caves, and shallow subsurface structures. Seismic data originates from two main sources. The first source encompasses waves generated by events within the Earth's interior, such as EQs, magma movement, volcanic eruptions, and large landslides. The second source involves artificial waves generated by human activities on intentionally constructed surfaces or in controlled environments such as pools.

To capture seismic waves, instruments such as seismometers, hydrophones (in water), and accelerometers are utilized. Seismometers measure the velocity of these waves, while accelerometers measure the strong motion or acceleration. The velocity and acceleration of seismic waves depend on the elasticity and density of the Earth's crust, mantle, and core. As depth increases, the velocity of seismic waves also increases. In the crust layer, velocities range from 2 to 8 km/s, while in the deep mantle, they can reach up to 13 km/s.

The accurate determination of the epicenter and hypocenter of an EQ source heavily relies on seismic networks that utilize multiple stations. The epicenter provides the geographical coordinates of the EQ's source. On the other hand, the hypocenter contains its depth. EQs are categorized as local, regional, or teleseismic based on their distances from the recording stations. Local EQs occur closest to the recording stations, followed by regional EQs, and teleseismic EQs occur farthest away.

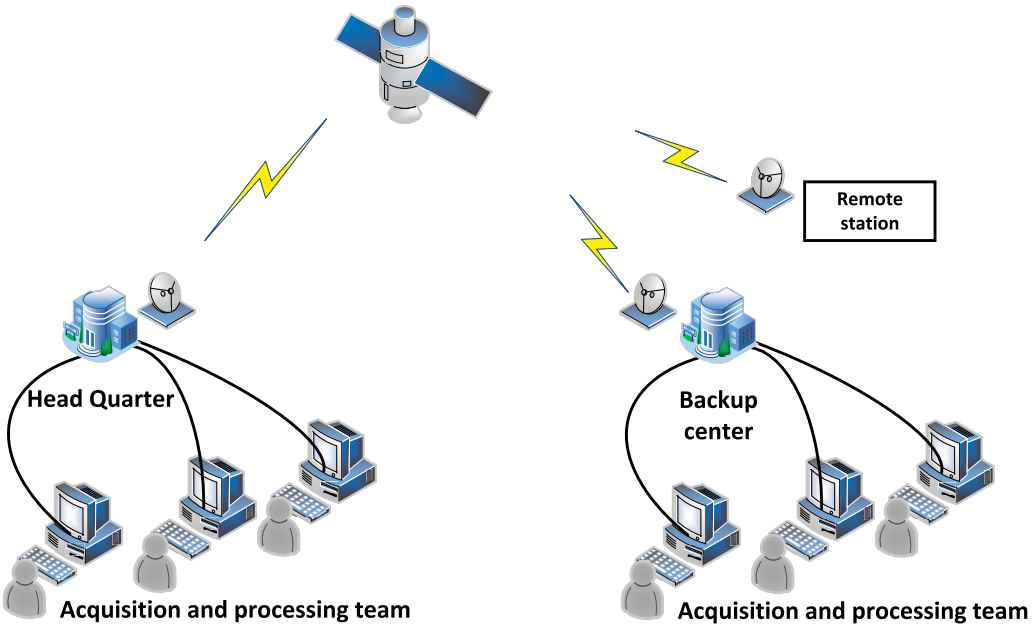


Fig. 3. Diagram of the general seismic observation system.

Table 2
Comparison of seismic network equipment.

Equipment	Sensor Type	Data Resolution	Deployment Depth	Communication Interface
Seismometers	Broadband, Short-Period	High	Variable (Surface to Borehole)	Analog/Digital, Ethernet
Accelerometers	Triaxial Accelerometers	High	Surface Installation	Digital (e.g., CAN bus)
GPS Receivers	GPS Timing Units	Sub-millisecond	Surface Installation	NMEA, PPS
Data Loggers	Digital Recorders	Variable	Surface to Borehole	Ethernet, Serial
Borehole Sensors	Borehole Seismometers	High	Borehole Installation	Analog/Digital, Ethernet
Power Supply Systems	Solar Panels, Batteries	Variable	N/A (Surface Installation)	N/A
Communication Modules	Wired/Wireless Communication	N/A	N/A	Ethernet, Wi-Fi, Satellite
Control Units	Central Processing Units	Variable	N/A	Ethernet, Serial

Seismic wave identification and analysis methods that have been around for a while use time-series analysis techniques like AIC, STA, and LTA. [46,47,48].[1] The LTA detects the slow-rising trend of seismic signal energy, whereas the STA detects sudden spikes. An indicator of signal quality is the signal-to-noise ratio, which is defined as STA divided by LTA. If the SNR surpasses a specific threshold, it indicates the presence of an EQ event. Upon detection, seismometers at multiple stations are used to calculate the hypocenter. Parameters such as the distance between receivers and the distance mean corresponding to the utmost active seismic zone are both factors that impact the determination of time window widths for STA and LTA. The duration of the set of multiple identical seismic signals is less than that of the set of multiple identical seismic noise variations, despite the fact that the STA period is longer than the LTA period. Using AIC, one aspect of the observed event is analyzed to determine the exact time of occurrence. Three components typically make up observed seismic waves, and the beginning time of the seismic wave is often determined using AIC [49].

In a research study presented in [50], a method for detecting listric faults is introduced with the aim of identifying potential risks associated with petroleum exploration in the Earth's crust. The methodology proposed by the researchers for interpretation is predicated on the texture of principal listric faults within a three-dimensional migrating seismic volume. This is achieved through the computation of a three-dimensional texture gradient, which illustrates the contrast between adjacent cubes along depth or time, inline directions, and crossline for each voxel in the seismic volume. The 3D volume map is then subjected to an adaptive global threshold, which generates a binary map displaying potential listric fault boundary locations. Additionally, image processing techniques described in [51,52] are employed to interpret seismic structures, thereby facilitating the identification of faults, salt domes, and other geological structures.

Seismic networks operate continuously, functioning 24/7, to monitor seismic waves in real-time applications [46]. These networks rely on various elements, mechanisms, and protocols, including the DCN and IoT, which facilitate data observation, transmission, and storage [47]. The unprocessed data is filtered and analyzed after its acquisition in order to identify seismic event parameters, including magnitude, depth, latitude, and longitude. [48]. Precise P-wave and S-wave arrival detection from different stations is crucial for this analysis. It is worth noting that the accuracy of event parameter calculations improves as the number of observing stations for the same event increases [49].

The system diagram depicted in Fig. 3 illustrates the configuration used for observing seismic data and transmitting it to the data center for analysis through RS tools [50]. This system comprises both hardware and software components. The team responsible for acquiring and processing seismic signals should possess adequate training and experience in working with such a complex system. Their tasks involve evaluating the quality of observed seismic data and performing essential processing steps such as picking, denoising, and discrimination. RS, AI, IoT, and optimization techniques all contribute significantly to seismic data analysis, hazard assessment, and DM [53,47,51,52,54,55,56].

[1] AIC = Akaike information criterion, STA = short-term average, LTA = long-term average.

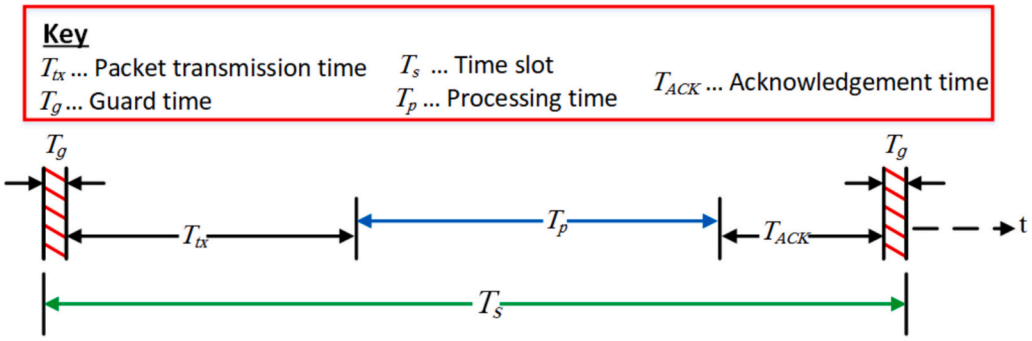


Fig. 4. TDMA slots distribution.

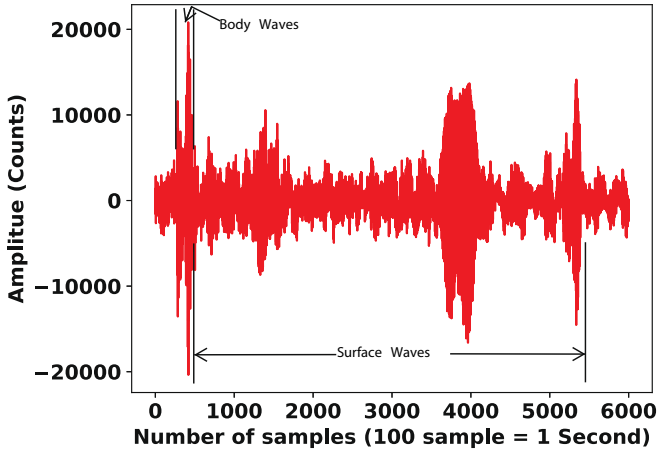


Fig. 5. Body Waves and Surface Waves.

2.1. Hardware elements

The system consists of various hardware components that perform different functions, such as seismometers, accelerometers, indoor modems, outdoor modems, and digitizers. Table 2 provides a list of these components along with brief descriptions. The first hardware equipment listed in the table is the seismometer, which is an indoor/outdoor device. It serves as a converter from mechanical wave to analog one (velocity wave), allowing for the transmission and reception of seismic data.

Similarly, the second piece of equipment is the accelerometer. It is specifically designed for measuring EQ acceleration. Accelerometers are essential for capturing the ground motion caused by seismic events. It offers high sensitivity and accuracy in measuring the acceleration associated with EQs, providing valuable information for seismic hazard analysis and structural engineering. Then, GPS is used to support the location and time. There is also a digitization process that enables the subsequent processing and analysis of seismic data using computational algorithms and software.

After observing the seismic data, we have to modulate/demodulate it via indoor or outdoor modules. It serves as a transceiver, facilitating the wireless transmission and reception of seismic data in indoor or outdoor environments. These modems are designed to withstand various weather conditions and provide reliable communication between different components of the seismic monitoring network.

In summary, the hardware equipment listed in Table 2 includes modems for data transmission and reception, a digitizer for converting analog signals to digital, an accelerometer for measuring EQ acceleration, and a seismometer for measuring seismic wave velocity. These instruments are essential components of a seismic monitoring system, enabling the collection and analysis of seismic data for various applications in EQ research, hazard assessment, and infrastructure monitoring.

Seismic networks primarily serve the purpose of monitoring seismic activities, including local, regional, and teleseismic events [46]. Local events refer to those that transpire within the country's geographical boundaries, while regional events occur outside the country but within the same continent. Teleseismic events, on the other hand, transpire on a different continent, far from the country in question [46]. The infrastructure of seismic networks commonly relies on leased lines, satellite communication, and LOS-based signals [46]. These techniques are commonly used to make seismic networks more data-friendly.

2.2. Software elements

The second wing for seismic monitoring is the software element. This wing represents a vital role in data gathering, data processing, and data analysis. The software is responsible for figuring out the wanted signal and discriminating it from noise. There is no doubt that this investigation process is also required to be done via human analysts in many cases. In fact, determining the gaps or data loss is an essential operation that is investigated by software. In other words, it looks for the incomplete observed data from the different seismic stations to achieve accurate data processing. In this process, several components are involved in the whole communication system utilized in the seismic network. Afterward, a process of trustworthiness and secrecy of the received data is executed to make sure that data can fill the gaps or missing stream.

It is worth mentioning that the data-retrieving process should use an accurate technique to organize the data transmission or retransmission. In other words, it is called a media access technique such as time division multiple access (TDMA). To this end, TDMA is used to divide the spectrum into a sequence of time slots. These slots are assigned for each seismic station to exploit for sending data through. Accordingly, such a technique facilitates the missing data investigation process in which it can synchronize between every transmitting station and the observatory. Also, it can keep every station transmitting in only its time slot. Therefore, it is easy to detect the lost data and from which station.

To make it clear, Fig. 4 illustrates how the TDMA mechanism operates showing each element in the process of transmission or retrieving lost data. The figure shows that every operation has its time window, i.e., time for packet transmission, time for acknowledgment, time for processing, and time for guard. More concretely, TDMA relies on a sequence number for every transmitted packet to make it easy to handle and retrieve if lost.

To sum up, the software is involved in data gathering, processing, and analysis, which leads to a robust and reliable seismic monitoring system. They coordinate with other parts of the network to find and get any missing data, and they use methods like time division multiplexing to get the signals that were detected back in sync. The precision and dependability of seismic data used in EQ monitoring, hazard assessment, and scientific research depend on these software functions.

Table 3
Summary of seismic signal processing techniques.

Seismic Signal Processing Techniques	Seismic signal processing techniques aim to enhance seismic signals, extract information about subsurface geology, and improve data interpretation.
Filtering	Enhances signal quality and reduces noise through frequency manipulation using band-pass, notch, low-pass, and high-pass filters.
Deconvolution	Removes wavelet effects and enhances data resolution for identifying subsurface structures and seismic events.
Migration	Relocates seismic reflections to their correct positions, compensating for wave propagation effects and improving subsurface imaging accuracy.
Velocity Analysis	Estimates subsurface layer velocities for accurate imaging and data interpretation by comparing observed and synthetic data.
Seismic Inversion	Mathematically models seismic data to estimate subsurface rock properties, such as acoustic impedance or density.

3. Seismic waves and seismic signal processing

3.1. Seismic waves

Geological events, volcanic eruptions, and man-made activities like mining and explosions all contribute to the creation of seismic waves, which are vibrations that travel through the Earth. The origins and interior structure of the Earth can be inferred from these waves. There are two primary categories of seismic waves (Body Waves and Surface Waves as shown in Fig. 5) [46,47]:

1. Body Waves:

- P-Waves: P-waves are the fastest seismic waves and are the 1st to arrive at a seismometer after an EQ. They are compressional waves that propagate through solid and liquid materials. P-waves cause particles in the medium to move parallel to the direction of wave propagation.
- S-waves: S-waves are slower than P-waves and arrive at seismometers after P-waves. They are shear waves that propagate only through solid materials. S-waves cause particles in the medium to move perpendicular to the direction of wave propagation.

2. Surface Waves:

- Love Waves: Love waves are surface waves that propagate horizontally and cause particles in the ground to move side-to-side in a horizontal motion. They are guided by the Earth's surface and have a significant impact on the damage caused by EQs.
- Rayleigh Waves: Rayleigh waves are surface waves that have a rolling motion similar to ocean waves. They cause particles in the ground to move both vertically and horizontally in an elliptical path. On the one hand, P-waves and S-waves are faster than Rayleigh waves. On the other hand, Rayleigh waves produce the most significant ground motion during an EQ.

3.2. Seismic signal processing techniques

Seismic signal processing techniques are used to analyze and interpret seismic data obtained from seismometers [57,58]. These techniques aim to enhance the quality of seismic signals, identify the characteristics of seismic events, and extract valuable information about the subsurface geology. Some commonly used seismic signal processing techniques include filtering, deconvolution, migration, velocity analysis, and seismic inversion [59,60,61,5,62,62,63,64]:

1. First: Enhancing the signal of interest in seismic data and eliminating noise requires the use of filtering [65]. This mathematical approach modifies the frequency component of the seismic data. When processing seismic signals, filters such as LPF, HPF, and BPF are frequently used. Filtering can be used to increase SNR and seismic event detection.
2. Second: By removing the influence of the seismic wavelet, a technique called deconvolution raises the resolution of seismic data [66]. By dividing the recorded seismic signal by the source wavelet, the reflection coefficient series may be estimated. Deconvolution contributes to the temporal resolution of seismic data improvement, better identification of seismic events, and to clarify the underlying structures.
3. Third: To create an accurate map of the subsurface, seismic reflections must be moved through the process of migration [67]. It assists in accurate subsurface imaging corrected by intricate wave propagation effects. Moreover, using subsurface velocity information and recorded seismic data, migration algorithms enhance the depiction of subsurface structures.
4. Fourth: Velocity analysis is employed to calculate the velocities of underlying layers, which is a crucial component of seismic processing [68]. Indeed, proper velocity data is necessary to interpret seismic data and the subsurface structure. It is standard procedure to compare synthetic data using different velocity models with the actual seismic data while doing a velocity study. Importantly, the velocity model is iteratively modified until the best-fitting model is obtained.
5. Fifth: The seismic inversion technique is used to measure the density and acoustic impedance of rocks that are located below the surface, which yields quantitative subsurface data [69]. More particularly, this technique aims to extract certain subsurface features by inverting seismic data using a range of algorithms and restrictions.

These seismic signal processing techniques play a crucial role in extracting valuable information from seismic data, enabling scientists and geologists to understand the Earth's subsurface structure, detect and locate EQs, and assess potential hazards associated with seismic events. Table 3 provides a summary of various seismic signal processing techniques. These techniques are used to enhance seismic signals, extract information about subsurface geology, and improve data interpretation.

4. Evolution of seismometers

Seismometers are instruments used to detect and record seismic waves. Over time, seismometers have evolved in design and technology, leading to significant advancements in the field of seismology. Fig. 6 illustrates the development of the seismometers' generations. The following subsection highlights the key stages in the evolution of seismometers:

4.1. Early seismometers

The earliest seismometers date back to ancient times and were simple devices that relied on human observations. These early seismometers, such as the Chinese seismoscope invented by Zhang Heng in the 2nd century, consisted of a suspended pendulum-like mechanism that would swing and trigger the release of a ball into one of eight dragon-mouth-shaped containers, indicating the direction of the EQ.

4.2. The invention of the modern seismograph

John Milne and E. H. Ambrose developed the current seismograph at the end of the nineteenth century. In the 1880s, Milne made a huge step forward in seismometer design with the invention of his horizontal pendulum seismograph. A hefty object was dangling from a horizontal beam; a pen was fastened to the object so that it could record the

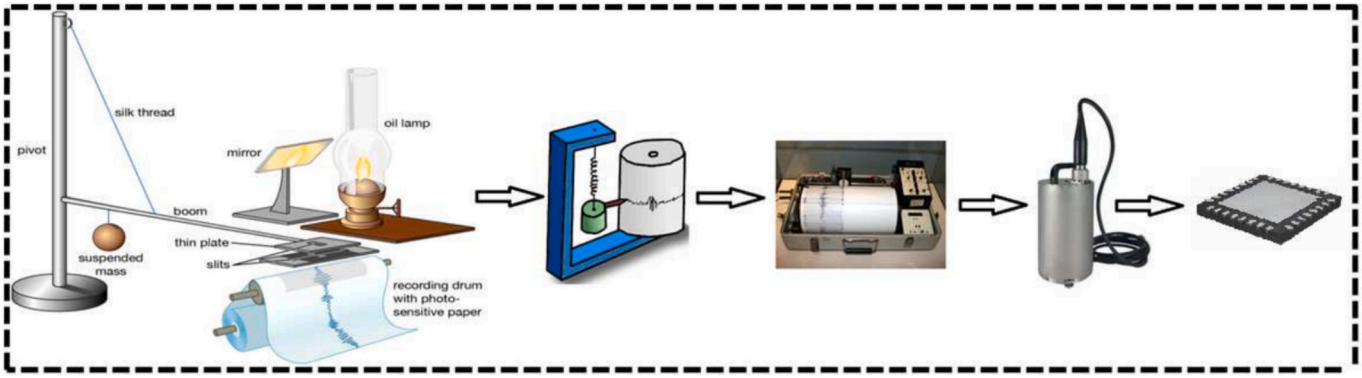


Fig. 6. Evolution of Seismometers.

Table 4
Evolution of seismometers.

Stage	Description	History	Sensitivity	Frequency Response	Size
Early Seismometers	Simple devices relying on human observations, such as the Chinese seismoscope [70,71,72].	132 CE	Limited	Limited	Large
The Invention of the Modern Seismograph	John Milne's horizontal pendulum seismograph revolutionized seismometer design, allowing for accurate measurement of vertical and horizontal ground motion [73,74].	Late 19th century	Moderate	Limited	Large
Introduction of Mass Spring Seismometers	Mass spring seismometers improved sensitivity and accuracy by utilizing a mass suspended by springs and a damping mechanism [75,76,77].	late 19th century	Moderate	Limited	Large
Development of Modern Seismometer Technologies	Introduction of electronic seismometers, followed by digital seismometers, offering improved sensitivity, dynamic range, and data storage capabilities [78,79].	Mid-20th century	High	Wide	Compact
Broadband and Strong-Motion Seismometers	Broadband seismometers expanded the frequency range of recorded seismic signals, while strong-motion seismometers capture high-amplitude ground motions during large EQs. [80,81,82].	Late 20th century	High	Broad	Compact
Advancements in Sensor Technology	Smaller and Cheaper devices used for EQ monitoring and structural analysis (MEMS-based seismometers) [83,84,85].	Recent	Moderate	Broad	Extremely compact

ground's movement on a paper-covered, spinning drum. Because of its layout, the ground's vertical and horizontal motion could be precisely measured.

4.3. Introduction of mass spring seismometers

The advent of mass-spring seismometers in the early 1900s allowed for more precise and sensitive seismic observations. In order to minimize oscillations, these seismometers used a mass that was hung by springs and a damping mechanism. The mass-spring seismometers were able to detect smaller EQs and record seismic waves more precisely.

4.4. Development of modern seismometer technologies

With the advent of digital and electronic systems in the middle of the twentieth century, seismometer technology experienced tremendous improvements. Modern electronic seismometers record and analyze ground motion by converting it into electrical impulses, replacing older analog seismometers. The data storage, dynamic range, and sensitivity of these electronic seismometers were all significantly enhanced.

The use of digital seismometers has increased in recent years. They enable remote monitoring and real-time data transfer through the use of sophisticated digital signal processing methods. Digital seismometers allow for the collection of massive volumes of seismic data for in-depth study and produce recordings with a high degree of detail.

4.5. Broadband and strong-motion seismometers

Broadband seismometers, developed in the late 20th century, expanded the frequency range of recorded seismic signals. They are capable of capturing a wide range of frequencies, from very low-frequency signals associated with large EQs to high-frequency signals associated with smaller local EQs.

Strong-motion seismometers, on the other hand, are designed to measure and record high-amplitude ground motions during large EQs. These seismometers are crucial for understanding the behavior and impact of strong ground shaking on structures and infrastructure.

4.6. Advancements in sensor technology

In recent years, advancements in sensor technology, such as the use of MEMS,[2] have led to the development of smaller, more compact, and cost-effective seismometers. MEMS-based seismometers integrate microfabricated accelerometers to measure ground motion accurately. These compact seismometers have found applications in various fields, including EEWS, structural monitoring, and portable seismic networks.

The evolution of seismometers has contributed significantly in advancing our understanding of seismic waves and EQ processes. Modern seismometers are characterized by enhanced sensitivity, accuracy, and digital capabilities. Thanks to these features, seismometers have revolutionized seismology and allowed scientists to detect and monitor EQs with greater precision. Table 4 provides a summary of the evolution of seismometers, and the instruments used to detect and record seismic waves.

4.7. Seismometers using fiber optic cable

Recently, optical fiber has been used to observe EQ waves via a

[2] MEMS = micro-electro-mechanical systems.

promising methodology called distributed acoustic sensing (DAS) [86]. This methodology utilizes the entire optical fiber length representing a seismic sensor in which the backscattered light waves are analyzed to determine the EQ location. This new system is advantageous as it has very high immunity against noise, and can be deployed for very long distances. Furthermore, it can be employed in highways, railways, or pipelines. Accordingly, such a system is beneficial for the areas that suffer from a lack of instrumentation or hazardous ones.

In DAS, the SNR poses a primary concern, as the backscattered light signals are weaker compared to traditional seismometers, making it more challenging to detect weak seismic signals, particularly in noisy environments. Additionally, spatial distortions along the fiber's length due to variations in construction require careful calibration. Fiber optic cables are also sensitive to environmental factors like temperature and strain, necessitating ongoing monitoring and calibration to mitigate measurement errors. Nevertheless, ongoing research aims to address these limitations, and as fiber optic-based seismic monitoring systems evolve, they are expected to significantly contribute to advancing EQ research, hazard assessment, and the development of EEWS.

Acoustic sensors, particularly DAS techniques, have emerged as powerful tools in seismology for capturing dense networks using fiber cables. Indeed, this approach is cost-effective and enables the time-lapse imaging of shear wave velocity and combines multichannel analysis of surface waves with ambient noise interferometry [87].

A prominent study carried out by researchers [87] utilized a set of linear DAS to continually track noise for three weeks in a row. Subtle changes, such as changes in water substance, near the surface could be observed using this method. Insights into the subsurface's dynamic behavior and characteristics were obtained by analyzing the recorded data; these insights are critical for understanding geological processes, monitoring groundwater resources, and evaluating the stability of structures.

When it comes to providing high-resolution measurements over expansive geographic regions, DAS techniques stand out. Dispersed antenna systems (DAS) use readily connected fiber-optic cables to convert preexisting infrastructure, such as telecommunication networks or oil and gas pipelines, into dispersed sensors, as opposed to conventional seismic arrays, which require the deployment of multiple individual sensors. Not only does this cut prices, but it also eliminates logistical hurdles. DAS technology enables long-distance, continuous seismic activity monitoring, which improves EWSs, detects small changes in ground motion, and provides academics with a wealth of data for event characterization. We can learn more about EQ behavior and geological processes thanks to the continuous data capture, which allows us to analyze trends over time and identify seismic patterns.

5. Remote sensing, autonomous robots and drones in seismology

5.1. Remote sensing

RS is an advancing technology with significant applications in seismology, encompassing terrestrial devices and satellite systems [88]. Its utility extends to various research endeavors, including detecting thermal activity, identifying ozone anomalies preceding EQs, and studying magnetic field anomalies in the ionosphere linked to powerful EQs. [88].

In a study conducted by Jing et al. [89], the authors examined tropospheric ozone before and after the 2008 Wenchuan EQ. They analyzed ozone volume mixing ratios from atmospheric infrared sounder data at different pressure levels to identify distinct behaviors associated with severe EQs. The authors observed a significant increase in tropospheric ozone 5 days prior to the heavy event, particularly surrounding the Longmenshan fault zone. They considered factors like focal depth, focal mechanism, tectonic and geological environment, and meteorological conditions to explain the abnormal behavior of tropospheric ozone. Similar observations were made for the EQs of Lushan and Jiuzhaigou, which occurred in 2013 and 2017, respectively, indicating that elevated tropospheric ozone can serve as an indicator of EQ epicenters. The study also found that the magnitude of the EQ played a key role in the emergence of abnormal tropospheric ozone [89].

For landslide mapping, Lv et al. [90] proposed a method that utilized extremely high-resolution RS images to create a landslide inventory map. The method was found to be effective when tested on actual datasets of landslide sites. Unlike traditional methods, this approach leveraged bitemporal very high-resolution images without generating a magnitude change map [90].

Liu et al. [91] demonstrated the use of nighttime light imagery from the Defense Meteorological Satellite Program's Operational Linescan System for analyzing EQ resilience at the county level. They developed a model considering the EQ stability using a light index at nighttime and analyzed the static and dynamic adaptability of various counties. The results showed that in the intensity zone, extremely damaged counties in the Wenchuan EQ zone showed more static flexibility as one moved north to south. In the near run, counties in the plains and protoplasmatic regions recovered economically more quickly than counties in the mountains and plateaus. Most heavily damaged counties, except for Pingwu, Wenchuan, and Dujiangyan had returned to normal conditions by 2011, while 52 % of the generally affected counties achieved the same in less than three years. The study also identified socioeconomic factors such as population age distribution, social welfare, and industrial structure as potential indicators of soil liquefaction danger, land usage, average elevation, and EQ resilience [91].

Passive microwave RS has been employed to detect temperature anomalies associated with strong EQs. [92,93]. In particular, the study by the authors in [92] proposed a new technique called spatiotemporally weighted about the 1st law of geography and its extension in the time domain to recover microwave brightness temperature anomalies. By employing nonseismic data, a spatially weighted background and a temporally weighted background were calculated, with the former being determined by the Euclidean geographical distance from the epicenter and the latter by local meteorological disturbance and long-term trends, respectively. In order to statistically process data on outgoing longwave radiation and examine how tidal oscillations affect radiation changes both before and after EQs, the researchers in the paper [93] used the background field analysis approach. Nonseismic and distantly related fault structure areas showed discrete temporal distribution of anomalies and longwave radiation enhancement zones, according to the findings [93].

These cases show how RS has helped seismologists a lot and how it can help us understand EQ-related events even more. Improved seismic event monitoring and analysis made possible by data collected from satellite systems, aerial photos, and microwave RS methods have yielded important insights for EQ prediction, risk assessment, and mitigation plans.

5.2. Autonomous robots and drones

According to [94], autonomous robots are machines that can carry out tasks independently of humans. Natural disaster management (DM) allows for the utilization of autonomous robots for search and rescue missions in hazardous regions that people are unable to reach [95,96]. According to [97], these robots can gather information about the disaster, including where survivors are and how much damage has been done to the infrastructure, by using sensors and cameras. According to [98], autonomous robots can serve as a means of transporting food, water, and medical aid to places that are inaccessible by conventional paths. These autonomous machines may be instructed to attain their goal even when faced with debris and other impediments. The utilization of autonomous robots in building and structural safety assessments can also aid decision-makers by revealing which places are OK to enter and which ones should be avoided [99]. In order to detect any threats, such as gas leaks and other dangerous substances, autonomous robots

Table 5
Comparison of emergency communication network techniques.

Technology	Reliability	Coverage	Latency	Scalability
Landline Phones	High	Local	Low	High
Cellular Networks	Moderate to High	Regional to Global	Moderate	High
Satellite Communication	High	Global	Moderate	Moderate
RF	Moderate	Local to Regional	Low	Moderate
Internet-based Platforms	Variable	Global	Variable	High
Mesh Networks	Variable	Local	Variable	Variable
First Responder Net.	High	National	Low	High
Social Media Platforms	Variable	Global	Variable	High

can also do surveillance [100]. Typically, autonomous robots can greatly improve response efforts during natural disasters by providing crucial information and carrying out dangerous activities that humans are not fit to accomplish [101].

Autonomous and commanded from a distance, drones are a kind of unmanned aerial vehicle (UAV). Drones come in a variety of styles, and each one has its own set of features and capabilities. One type is the fixed-wing drone, which is designed to fly long distances and stay in the air for extended periods of time. They are typically used for mapping, surveying, and monitoring large areas. Another type is the rotary-wing drone, which uses a set of rotors to hover and move in any direction. These drones are commonly used in photography, cinematography, and search and rescue operations. Hybrid drones combine the features of both fixed-wing and rotary-wing drones, allowing them to take off and land vertically like a rotary-wing drone, but also fly long distances like a fixed-wing drone. In the end, there are drones with just one rotor blade. These are the ones you'll usually see flying about commercial and industrial areas, checking things like power lines and wind turbines.

Using a variety of communication methods, drones are able to converse with a wide range of entities, including users, ground stations, satellites, and even other drones. Using radio communication frequencies like Wi-Fi or Bluetooth is a typical way for drones to talk to their ground stations or users. Because of this, the drone can be controlled and data, such as live video feeds, may be transmitted in real-time. According to [102], drones are able to communicate with satellites through Satcom or the Global Navigation Satellite System, allowing for long-range communication and accurate positioning. Furthermore, drones are capable of exchanging data with one another through mesh networking, in which drones function as nodes, or by utilizing specific communication protocols such as Drone-to-Drone Communication (D2D) or Vehicle-to-Vehicle (V2V) communication [103]. According to [104], drones may communicate in several ways, which allows them to fly independently, work together with other drones, and send data to people on the ground in real-time [105].

Natural DM may make use of drones to provide precise and real-time data on the extent and severity of a natural disaster, along with its impacts on infrastructure and the environment [106]. According to [107], drones can be utilized for surveying inaccessible or hazardous locations, including those impacted by natural disasters like landslides, floods, or wildfires. To help decision-makers allocate resources and prioritize response operations, drones can map and simulate the extent of damage caused by a natural disaster. Supplying areas that are inaccessible by traditional means of transportation with necessities such as food, water, and medical aid is another potential use of drones [98].

5.3. Transmission of real-time monitoring data

In seismology, the transmission of real-time monitoring data is essential for the rapid and accurate analysis of seismic occurrences. In order to receive and process vital seismic data, it is common practice to link all seismic stations to a central processing facility using a specialized communications method. Nevertheless, depending on just one communication backhaul technique is risky, especially when dealing with catastrophes that can damage network infrastructure [108].

Enhanced robustness in the transmission of real-time monitoring data has been emphasized by seismologists and emergency response teams in recognition of the network infrastructure's susceptibility. Incorporating various forms of backhaul into an emergency communication network is one way to tackle this difficulty. To lessen the impact of any one point of failure, and increase the system's resilience and dependability, it is recommended to diversify the communication infrastructure.

Various groupings of communication network strategies are compared using various criteria in Table 5. A few examples of these requirements are application coverage, availability of preexisting infrastructure, self-organization capabilities in partially damaged network infrastructure, deployment cost, and survival in emergency situations. If we want to build a communication network that can survive disasters and keep sending real-time monitoring data, we must evaluate these criteria.

What we mean by “application coverage” is how well the communication network can link seismic stations all across a certain region. To make sure all the important monitoring stations are properly linked, this criterion takes the network's range and scalability into account. Another important factor is the communication network's capacity to survive and recover from interruptions induced by calamities like EQs or extreme weather events [109].

In addition, the feasibility and simplicity of establishing a communication network are heavily influenced by the presence of existing infrastructure. Saving money and time during deployment may be possible by making use of preexisting infrastructure like cellular networks or satellite links.

Seismologists and emergency response teams can make educated decisions about building a dependable communication infrastructure for transmitting data from real-time monitoring by taking these parameters into account and comparing various communication network strategies. The objective is to guarantee the timely delivery of critical seismic data to the processing facility so that decisions may be made effectively and in a timely manner during seismic occurrences.

6. Social networks and IoT for seismology

6.1. Social networks for seismology

In the field of seismology, social networks have grown into vital resources for making decisions and, in the event of a natural disaster, for perhaps rescuing lives. Experts in the field have acknowledged the influence of social media and have pushed for all-encompassing strategies that utilize these sites to lessen the negative impact of catastrophes [110].

Recent research has investigated how social media platforms, particularly Twitter, can be used to collect information regarding the condition of infrastructure in the case of a seismic event [111]. Anomalies in Twitter activity have been identified and links between the amplitude of an EQ and the volume of associated tweets have been established by researchers by analyzing user actions and patterns of Twitter activity during EQs. One study that looked at EQs from 2018 found that major events prompted more than four million tweets, which shows that social media could be a sign of seismic activity. However, it is important to note that the intensity of the EQ and population density also play significant roles in determining the level of social media activity [111].

The role of social media extends beyond data gathering. Researchers have explored how social networks can facilitate feedback and relief administrations during and after seismic events [112]. The transmission of warning messages through social media platforms has also been investigated as an effective means of communication during the early

Table 6
Benefits of social networks during earthquakes.

Benefit	Description
Real-Time Updates	Provides immediate updates on earthquake occurrences, helping monitor the situation and inform the public.
Early Warnings	Facilitates rapid dissemination of early warning messages, allowing time for protective actions.
Feedback and Relief	Collects feedback from affected individuals, aiding in the coordination and administration of relief efforts.
Public Awareness	Enhances awareness about earthquake preparedness and response protocols.
Hazards Assessment	Organizations like USGS gather data via social media to assess seismic hazards.
Rescue Coordination	Helps coordinate rescue operations by sharing information about affected areas.
Damage Reports	Allows public reporting of damage, aiding authorities in mapping impact and prioritizing response.
Aid Distribution	Facilitates organization and distribution of resources by providing real-time needs information.
Vital Information	Share critical info about safe locations, shelters, and essential services.
Community Support	Provides a platform for affected individuals to connect, offer support, and share experiences.
Post-disaster recovery	aid in long-term recovery efforts by maintaining communication, offering continued support, and disseminating information about rebuilding and aid opportunities.

warning phase [113]. Social media has been employed to enhance public awareness and emergency response efforts, providing real-time updates and monitoring of natural disasters [114,115]. One of the worldwide organizations concerning the seismological aspects is the USGS. It has developed the use of short text messages and social media to improve gathering information for seismic hazards assessment [116]. Additionally, social networks can play a crucial role in the aftermath of earthquakes by facilitating the coordination of rescue and relief efforts. Platforms like Twitter and Facebook enable the rapid dissemination of information about affected areas, helping to direct emergency services and volunteers to where they are most needed. Crowdsourcing through social media can also help map damage and identify areas requiring immediate attention, significantly speeding up the response time. Furthermore, these platforms allow for the efficient distribution of resources and aid, as well as the sharing of vital information about safe locations, shelters, and essential services. Overall, the integration of social media into earthquake preparedness and response strategies can greatly enhance the effectiveness and timeliness of disaster management efforts.

Indeed, social networks have several drawbacks for elders and rural communities during natural disasters. Limited access to technology is a significant issue, as many elderly individuals do not have smartphones, computers, or internet access, and rural areas often suffer from poor internet connectivity and infrastructure, hindering timely access to information. Additionally, digital literacy is a major challenge; elders may struggle to use technology effectively, and rural populations typically have lower levels of digital literacy, making it difficult for them to navigate and utilize social media platforms. Another critical issue is the rapid spread of misinformation and fake news on social networks, which can mislead elders and rural residents during emergencies. The reliance on visual and text-based communication on social media can also be problematic, especially for individuals with visual or cognitive impairments, which are more common among the elderly. Furthermore, language barriers pose a challenge as social networks often cater to dominant languages, potentially excluding those who speak regional or minority languages prevalent in rural areas.

To address these drawbacks, several solutions can be implemented. Improving access to technology by investing in internet infrastructure in rural areas and providing subsidized devices to elders and rural residents is essential. Enhancing digital literacy through community-based training programs tailored for these groups and creating support networks with tech-savvy volunteers can help them use social media more effectively and safely. Combating misinformation requires promoting the use of official channels and trusted sources, as well as employing community moderators to monitor and correct false information. Additionally, alternative communication methods such as automated phone calls or voice message systems can deliver critical information to those who may not use or understand social media. Leveraging traditional media like local radio and TV, which have high penetration in rural areas and among the elderly, can also be effective. Ensuring that social media platforms and emergency information are available in multiple languages, including regional and minority languages, and providing translation services for critical information, can help overcome language barriers. By implementing these solutions, the efficacy of social networks during natural disasters can be significantly improved for elders and rural communities.

Table 6 shows the benefit of social networks for EQs for specifying events. In other words, the table presents how they interact to determine the events. The interaction includes some factors such as the number of posts, the sentiment expressed in social media discussions, the geographical distribution of social media activity, and other relevant parameters. More particularly, social media can be used to track the EQ and the people's interactions due to the EQ disaster. Accordingly, the analysis of these factors can offer valuable information for seismologists and decision-makers about the EQ and its impact, and how to perform an efficient DM plan. This plan can be categorized as during and post-disaster.

6.2. *Internet of things networks for seismology*

Recently, the role of EEWS significant role in DM has emerged, where an IoT system can facilitate monitoring the afflicted areas by an EQ and can support a fast alarm. Accordingly, IoT can contribute to the DM and alleviate EQ consequences. Much study has gone into the potential of software-defined networks (SDNs) [117], network functions virtualization (NFV), DCNs, RFID, 5G, and satellite systems to improve EQ mitigation [118,105,119,120,121,122,123,307].

In addition to traditional approaches, social media platforms have also played a minor but important role in reducing the risks associated with EQs. These platforms enable the rapid dissemination of information and facilitate communication during and after seismic events. Recognizing the potential of combining the power of social media with advanced networks, researchers have proposed comprehensive approaches to mitigate the adverse effects of EQs. [110]. By leveraging the real-time information shared on social media platforms [124,125], authorities and emergency responders can gain valuable insights into the impact and extent of the EQ, enabling them to make informed decisions and allocate resources effectively.

The literature on IoT networks for seismology goes beyond RS supported by satellite observations [126]. Researchers have also explored the utilization of SDNs and virtualization, incorporating IoT gateways [127]. The objective of these endeavors is to assist large-scale networks where the infrastructures have been partially or completely destroyed by EQ events. By leveraging the flexibility and programmability of SDNs, combined with the virtualization capabilities of network function virtualization, IoT networks can be rapidly deployed and dynamically reconfigured to provide critical communication and sensing capabilities in disaster-stricken areas. These technologies enable the establishment of resilient and adaptive networks that can quickly restore connectivity and support emergency response efforts.

Virtualization, in particular, has the potential to revolutionize the field of catastrophe risk reduction. In a study [128], researchers presented a catastrophic scenario that depicted an EEWS for safe evacuation from disaster risks. The system integrated cloud systems and heterogeneous virtual networks to provide real-time information about potential hazards and enable timely and effective evacuation measures. By leveraging virtualization technologies, the system demonstrated the ability to dynamically allocate computing and communication

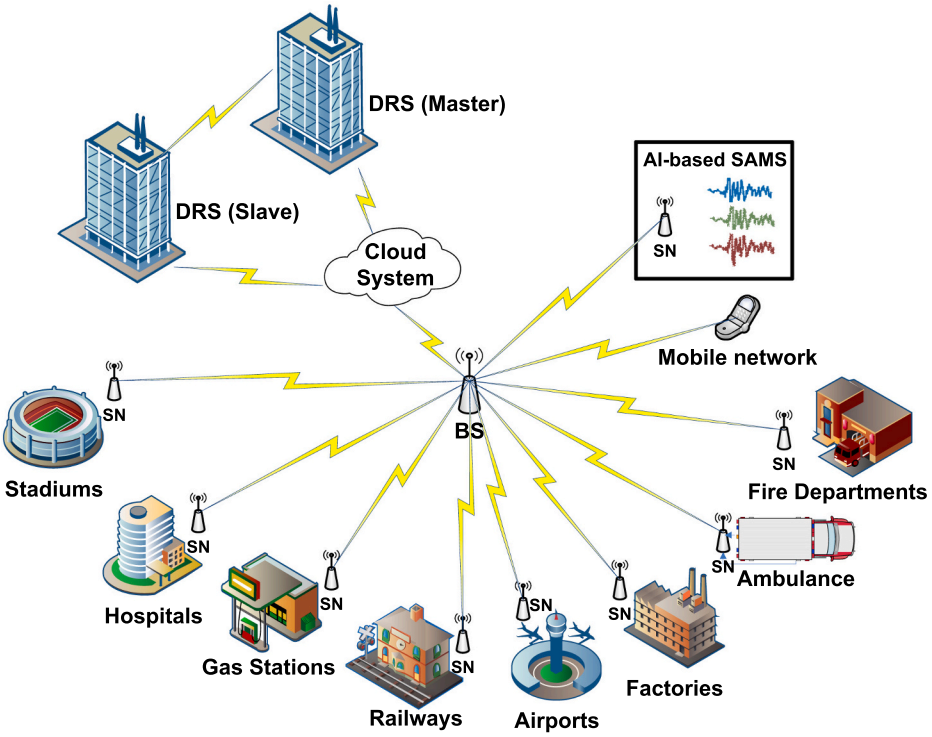


Fig. 7. EEW architecture based on IoT systems.

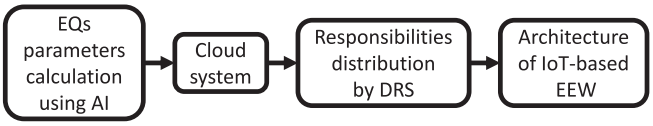


Fig. 8. Workflow of the general system.

resources, ensuring the seamless operation of the EEWS even in the face of infrastructure disruptions.

All of these initiatives are proposed in addition to the conventional techniques for EQ detection, research, and fault rupture type distinction, which were extensively studied in the literature [129,130]. In [131], The authors proposed an approach for EQ detection based on local similarity, which utilized the nearest neighbor method to assess the coherence of received signals between target stations and their immediate neighbors. EQ magnitude estimation has been studied within the first few seconds [131,132].

In fact, more investigation is required to tackle the temporal limitations linked to calculating EQ characteristics by traditional techniques. Irrespective of the selected approach, it is imperative to acknowledge that a sufficient period of delay typically elapses before the source time function matures. The precise approach used will determine the actual computation time. For EEWSs, the first few seconds are very important since they allow for the evaluation of the arrival time and wave height of the first inbound EQ wave, also referred to as the P-arrival wave. The amount of time that must pass before strong EQ waves arrive must be taken into account since this will have an immediate effect on the blind zone's width and the window size that is evaluated.

In the context of ecological catastrophe risk management, a data-driven core type from an EWS utilizing an IoT system has been proposed [133]. The authors in [134] discussed the effective exchange of information across multiple platforms and the use of IoT and D2D facilities for suitable data flow to prevent the spread of disasters. Additionally, EEW systems have utilized message queuing and web applications [135,136].

Fig. 7 illustrates an EEWS based on the IoT, as described by Abdalzaher et al. [18]. The system utilizes a real-time program called SAMS, which incorporates AI to automatically evaluate EQ activity.[3] The proposed workflow, as depicted in Fig. 8, begins with the trigger of an alert based on EQ characteristics and continues until the relevant organization takes appropriate measures to mitigate the risks associated with the EQ catastrophe.

In this system, the sink node is the one linking to the IoT cloud and facilitates communication between the SAMS system and the rest of the components. When the SAMS system senses the parameters of an EQ, the system promptly uses the IoT cloud to send an alarm message to the disaster recovery system (DRS) with the EQ parameters.

The DRS connects to all of the IoT system's components via the IoT cloud and decides what has to be done after getting the alarm message. In order to quickly respond to the EQ tragedy and lessen its effects, the DRS serves as the central hub that organizes the actions to be performed by every entity in the IoT system, such as trains, stadiums, companies, etc.

The DRS then informs each entity of what has to be done. Sensor nodes (SNs) are used to allow all system components to receive the action triggers from the cloud. IoT devices ought to be linked to these SNs. The Tmote Sky SN, which is commonly utilized in IoT-based networks built on the standard of IEEE 802.15.4, is one instance of such an SN. The Tmote Sky SN offers precise location information and various sizes suitable for IoT applications [137]. Table 7 presents a summary of the common IoT nodes that can sever the EQDM.

7. Optimization and seismology

Fundamentally, optimization refers to the process of maximizing benefits while minimizing costs. Specifically, in the context of a given problem, it involves representing the problem as a mathematical formula, known as the objective function. This function is subject to

[3] SAMS = seismic activity monitoring and analysis system.

Table 7
Comparison of wireless SNs for IoT applications.

Sensor type	MAC Protocol	Standard	Environment	Temp.	Founder	Protocol	Topology	Radio Chip type	Coded by	Application	Pros	Cons
Mica2	CSMA-CA	3G	In/Out door	-40°C to 85°C	CrossboW	No predefined protocol	Star and IoT paper	CC 1000	TinyOS	IoT [138]	Supports ad-hoc mesh networking	Intensive HW
Micaz	CSMA-CA	IEEE 802.15.4 and ISM band	In/Out door	-40°C to 85°C	CrossboW	No predefined protocol	Array topology [139]	CC 2420	TinyOS	IoT [138]	Supports ad-hoc mesh networking	Intensive HW
Tmote Sky	CSMA and TDMA	IEEE 802.15.4	In-Out door	-40°C to 85°C	University of California, Berkeley	Zigbee [140], 6LoWPAN [138]	Star, Mesh [138]	CC 2420	TinyOS	IoT [140]	Supports external flash	Intensive HW
SHIMMER	CSMA and TDMA	IEEE 802.15.1	In/Out door	-40°C to 80°C	Intel	Bluetooth Protocol	Tree [141]	CC 2420	TinyOS	Health monitoring [141] Magnetic, ECG, Galvanic Skin Response, PIR	Supports Bluetooth and micro SD card	Lack of formfactors
IRIS	CSMA-CA	IEEE 802.15.1 and ISM	In/Outdoor	-40°C to 85°C	Crossbow	Token ring [142]	Mesh [143]	Atmel	TinyOS	Temperature, humidity, light, soil moisture, wind speed and direction [144]	Supports high data rate	No Applicable sensors
EZ430-RF2480-F2500T	CSMA-CA	2.4GHz ZigBee	Indoor	-40°C to 85°C	Texas Instrument	Zigbee	All topologies	CC 2480	C	Bluetooth [143]	Supports USB interface and high data rate	High power consumption in Idle/sleep modes
Wasp-motes	CSMA-CA	IEEE 802.15.4	In/Outdoor	-40°C to 70°C	Libelium	LoRaWAN	Star, tree, mesh	CC 2500	C++	LoraWAN	Upgrade or change of firmware versions without physical access	-Switch-off modules before low-power mode -The outdoor performance << the indoor
SunSPOT	CSMA-CA	2G and 6G, IEEE 802.15.4	In/Outdoor	-40°C to 85°C	Sun Microsystems	Zigbee and LowPAN	Large scale magnetic topology [145,146]	CC 2420	J2ME	Solarium application, space and magnetic activities [145,146] Robotics, Art, Toys, Electronics	SW and HW are open source	Expensive

Table 8
Optimization techniques in seismology.

Optimization Technique	Application	Advantages	Challenges
Genetic Algorithms [153]	Inversion, Source Location	Global Search, Parallelism	Parameter Tuning
Simulated Annealing [154]	Velocity Model Inversion	Convergence to Global Minima	Temperature Control
Particle Swarm Optimization [155]	Travel Time Tomography	Rapid Convergence, Simplicity	Sensitivity to Initial Conditions
Ant Colony Optimization [156]	Seismic Source Localization	Robustness, Flexibility	Computational Intensity
Differential Evolution [157]	Inversion of Seismic Data	Population Diversity	Control Parameters
Genetic Programming [158]	Seismic Attribute Analysis	Adaptability, Expressiveness	Computational Cost
Tabu Search [159]	Seismic Waveform Inversion	Solution Space Exploration	Parameter Configuration
Harmony Search [160]	Joint Inversion of Geophysical Data	Simplicity, Robustness	Limited Exploration
Artificial Neural Networks [161]	Seismic Signal Processing	Nonlinear Mapping, Learning	Overfitting, Data Quality
ML Ensemble Methods [162]	EQ Prediction	Model Robustness, Generalization	Training Data Bias

constraints that define the search space for finding the optimal solution to the optimization problem.

When it comes to seismic signal analysis and interpretation, optimization techniques are crucial, especially in seismic inversion and seismological modeling. Optimization can be involved in the EQ parameters estimation such as peak ground acceleration (PGA), peak ground velocity (PGV), location, and magnitude. An essential step in the oil and gas industry's seismic inversion process is to use the reflected seismic data to quantitatively characterize the subsurface rock parameters, including bulk density, and sonic velocity via multi-objective optimization. In [147], the authors have studied four datasets (from the Gulf Coastal Plain of Texas, Southeast New Mexico, and West Texas, as well as the Middle East). It was concluded that using a multi-object optimization problem on multiple seismic parameters can achieve a better grasp of the subsurface structure.

In addition, differential semblance optimization has been utilized for the convergence and stability of seismic inversion [148]. More particularly, that study focused on the interchange between the reflection seismology estimation and the least-square inversion in seismograms. It was concluded that the main concern in that regard is the standard approximation inversion of plane waves in a layered constant density of the acoustic medium. In [149], the authors have developed a PSO model EQ intensity estimation in terms of measuring the PGA. An Iraqi database containing 187 recordings of historical ground motion from several tectonic zones in Iraq was used by the model. The PSO model improved the PGA estimate by taking seismic factors such as average shear-wave velocity, EQ magnitude, focal depth, and the nearest event to a station into account; this provided useful information for EQ intensity assessment.

Seismological modeling has also made use of optimization approaches. An optimization function was created to investigate diverse geological layers in a single study [150]. By optimizing the fit between observed seismic data and simulated models, this approach made it possible to identify and characterize subsurface structures. Furthermore, according to another study [151], seismological modeling could benefit from a multi-objective PSO that utilized Pareto optimization. Improved accuracy and resilience in seismological modeling were the results of this approach's ability to optimize many objectives simultaneously.

Interestingly, optimization has significantly contributed to prolonging the seismic station's lifetime. In [152], the authors have developed a mixed integer programming (MIP) model to optimize the packet length that would be transmitted, which is reflected in preserving the station power source. Moreover, the model optimized the inter-distance between communicating nodes, which is important for the dense seismic network relying on IoT nodes. This idea has been adopted in an IoT-based EEWS as proposed in [18]. Accordingly, it is not limited to prolonging seismic network lifetime but also contributes to preserving human life. Table 8 categorizes the optimization techniques in seismology.

8. AI for seismology

In the current era, AI represents a cutting-edge tool in seismology. It has provided fast, accurate, and reliable solutions for EQ wave analysis, parameter estimation, and providing risk reduction maps. AI systems can evaluate the risk and pinpoint areas vulnerable to them by sifting through mountains of seismic data [163]. Urban planners, infrastructure developers, and emergency planners can all benefit greatly from the data provided by these risk reduction maps. Stakeholders are empowered to make educated choices on building codes, land use rules, and evacuation routes, which in turn reduces the effect of EQs on human lives and infrastructure [164,165].

In addition, AI has been important in developing and executing methods to effectively mitigate threats. AI algorithms can forecast the events that are most likely to occur by looking at past seismic data and adding additional variables like geological features and tectonic plate movements. In order to prioritize and deploy resources where they are most needed, stakeholders can utilize this predictive capability [166]. For instance, AI can assist in determining which vital infrastructure, including schools, hospitals, and bridges, are more vulnerable during a disaster so that specific reinforcement and retrofitting can be done to make them more resilient.

The rise of DL as a state-of-the-art technology that can solve complicated problems without using specialized mathematical techniques has been well-documented [167,168]. It is suitable for mining and reconstructing irregular as well as regular data due to its capacity to learn complex patterns and characteristics from seismic data. To estimate partial seismic data with different patterns, including regular and irregular, DL models have been suggested in the seismology field [169,170]. Loss functions like mean squared error have been used to evaluate these models, and optimization approaches like the Adam method have been employed [171].

A taxonomy of research that has used ML and DL for particular aims within the area is presented in Table 9. The table showcases different seismology jobs that have been enhanced by artificial intelligence.

8.1. Earthquake detection

To detect seismic activity in the continuous dataset, the convolutional neural network (CNN) technique is used [227]. Using a 1000-sample window as the CNN's input is the key to this technique. In the binary representation of the final output, EQ windows are given the value 1 and noise windows the value 0. The CNN network takes the three parts of the seismograms as inputs and models the problem as a binary classification problem with two classes of outputs. Datasets used for training and testing were collected in the US state of Oklahoma. The strategy's efficacy stems from the utilization of many CNN layers, each meticulously built to extract distinct degrees of abstraction. The network's capacity to learn and distinguish between EQ and noise windows is improved as a result.

Conversely, converting seismic data from the time domain to the time-frequency domain is proposed as a method to improve the ability to identify EQs, as supported by prior research [228,229]. This strategy

Table 9
AI for seismology.

Target	Proposed paradigm	Evaluation	Work
Denoising	Denoising using Autoencoder (AE)	85.8% < Accuracy < 98.9%, MAE > 2.38×10^{-5}	[172,173]
	DNN	SNR-Based	[174]
	DeepTensorCNN	SNR-Based	[175]
	Dual-path RNN	Using SDR & SNR scale-invariant	[176]
Clustering	AP	Using SDR & SNR scale-invariant	[177]
	DCAE	91% < Accuracy < 98.5%	[178]
	HSVR	Via Hierarchical model	[179]
	A model using Deep scattering	Via Hierarchical model	[180]
Location	DAE with Convolutional Neural Networks	MSE → 0.0000515	[18]
	Convolutional Neural Networks	Accuracy → 74.6%	[181]
	DCNN	Location error < 300m	[182]
	Long Short-Term Memory-FCN	Accuracy → 89.1%	[183]
Magnitude	DAE with Convolutional Neural Networks	MSE → 0.000028	[18]
	Convolutional Neural Networks and RNN	SD ≈ 0.2, MSE ≈ 0	[184]
	Support Vector Machine	SD ≈ 0.33	[185]
Phase detection	Convolutional Neural Networks	Accuracy → 99.8%	[186]
	Convolutional Neural Networks with RNN	Accuracy > 98%	[187]
	PhaseNet software	Accuracy → 50% - 60%	[188]
Picking	Convolutional Neural Networks	Accuracy → 63.8%	[189]
	Convolutional Neural Networks	Accuracy → 92.5% - 98.6%	[190]
	Convolutional Neural Networks	Accuracy → 95%	[191]
	Capsule network	Accuracy → 97.64	[192]
	Convolutional Neural Networks with RNN	Accuracy > 98%	[187]
	Long Short-Term Memory	Accuracy → 98.8%	[193]
PPV	ANN	Accuracy → 95%-98.6%	[194,195]
	Support Vector Machine	Accuracy → 96%-99%	[196,197]
	FIS	Accuracy → 95%	[198]
	ANFIS	Accuracy → 98%	[199]
Strong motion wave	Decision Tree	Accuracy → 99.7%	[200]
	US manifold-approximation	-	[201]
	ensemble model	89.4% < ROC < 93.6%	[202]
	Gradient boosting	R2	[203]
Quarry-blast and EQ classification	ANFIS-GWO	Accuracy → 99.8%-99.9%	[162]
	Convolutional Neural Networks and RNN	Accuracy → 99%	[204]
	ANN	Accuracy → 89%	[205,206,207]
Estimating incomplete regular and irregular data	Manifold learning techniques	94.44% < Accuracy < 99.05%	[208]
	CNNs	-	[209]
	AE and CNN U-net	Accuracy → 71.7% - 99.7%	[210]
EEWS	DAE with Convolutional Neural Networks	0.0001 < MSE < 0.0000033	[18]
	Long Short-Term Memory	Missed alarm rate of 0%, a false alarm rate of 2.01%	[211]
	Convolutional Neural Networks	8.5km < location error < 4.7km	[212]
Landslide	Multitarget regression and Ncheck algorithm	MAE = 2.78 s, MAE = 27.81	[213]
	U-Net	84.17% < MIoU < 88.4%,	[214,215]
	Adaptive sampling and RF	AUC → 90.6%	[216]
	Convolutional Neural Networks with RF	AUC → 95%	[217]
	Deep CNN	81.17% < Accuracy < 99.42%, MIoU = 56%	[218,219,220,221]
	AE	74.9% <F1< 89.1%	[222]
	LandsNet	Accuracy → 86.89%	[223]
	Full CNN	Accuracy → 88.2% - 93.85%	[224]
	RF	Accuracy → 93.2%	[225]
	Support Vector Machine	Accuracy → 80% - 94%	[226]

utilizes the more extensive data present in the time-frequency domain to enhance the categorization process. Due to the unique frequency properties of seismic signals in contrast to background noise, applying this transformation helps deep learning models extract more distinct features, thus improving classification performance. The detection approach utilizes the Short-time Fourier Transform (STFT) [228] as input for the deep learning model called CRED. The CRED system employs a CNN to extract features, which are then reconstructed into labels using many layers of bidirectional long short-term memory (LSTM) units. Nevertheless, the STFT is known for its limited ability to accurately represent data, which has led researchers to investigate other transformations, such as continuous wavelet and Stockwell transforms, in order to improve time-frequency representations [229].

8.2. Phase picking

According to [230], a model called generalized phase detection (GPD) has proven to be effective in correctly identifying seismic phases by integrating a window-based approach with a convolutional neural network (CNN). The GPD model showed promising results in recognizing phases when trained on data from California. The noise window, P-wave window, and S-wave window all receive zero, one, and two outputs, respectively, from the GPD, which takes four-second three-component seismograms as input. An input window of 400 samples is required by the California dataset, which has a sampling rate of 100 Hz. Noteworthy, the P-wave arrival falls exactly at sample 200 inside the specified P-wave window, and the noise window is removed before its arrival. Similarly, according to [230], the exact moment when the S-wave arrives is exactly halfway through the 4-s interval that is associated with the S-wave. On the other hand, datasets from states other than California show uneven performance when using the GPD model. Thus, a Capsule Neural Network was proposed because of its remarkable ability to generalize.

The capsule neural network has numerous advantages compared to the conventional CNN: 1) It captures spatial information by incorporating extracted features, which improves the accuracy of classification. 2) The orientation of the object is taken into account, allowing for more flexibility in training with limited datasets. 3) Its architecture reduces

the reliance on pooling layers, which helps retain important input data information [231]. The capsule neural network is structured with multiple consecutive layers, specifically the convolutional layer, capsule layer, and digit caps. The primary function of the convolutional layer is to extract a wide range of feature maps from the input data. Afterward, the capsule layer arranges the retrieved feature maps into separate groupings known as capsules. However, there are other models that have been expressly designed to tackle the difficulty of phase-picking. PhaseNet, EqTransformer, and EQCCT [232,233] are prominent examples that are well-acknowledged for their ability to detect P and S-wave arrivals in EQs. PhaseNet utilizes a UNET structure and demonstrates excellent performance when trained on seismic data from Japan. EqTransformer utilizes sequential attention networks, whereas EQCCT leverages the benefits of the compact convolutional transformer (CCT) [233]. Nevertheless, EQCCT exhibits enhanced generalization capability when applied to diverse seismic datasets, as evidenced by its higher performance metrics in terms of both the quantity of missed events and accuracy in event detection.

8.3. Magnitude and location estimation

Estimating the magnitude and location of an EQ is crucial for comprehending seismic events since it indicates the amount of energy that is produced during these occurrences. The Richter scale is a commonly utilized method for quantifying the magnitude of an EQ. [234]. Conventional seismic measurement enables us to analyze and differentiate the effects of EQs on a global scale. The magnitude plays a crucial role in understanding the possible intensity of ground shaking, forecasting the susceptibility of infrastructure, and formulating robust seismic building regulations. Precise magnitude assessment is crucial in EEWS, as it ensures the reliability of alarms. The utilization of CNN and recurrent neural network (MagNet) has proven to be effective in estimating EQ magnitude, particularly by analyzing 60-s three-component seismograms [235]. The MagNet model was trained using the STEAD dataset, which consisted of over 300,000 seismic waveforms with epicentral distances of less than 1 degree. MagNet allocated 70 % of the extracted waveforms for training, 10 % for validation, and 20 % for testing during the training phase. MagNet has a unique characteristic where it does not standardize the input waveform, enabling the use of amplitude data during the training phase. The outcomes derived from the utilization of MagNet in estimating both local and duration magnitudes have demonstrated considerable potential, highlighting the effectiveness of this approach in EQ magnitude estimates.

Moreover, a novel method utilizes an attention network to estimate EQ magnitude by dynamically taking into account a changing group of stations [236]. That model employed a hybrid model using CNN and fully connected ones for magnitude estimation via different datasets from Chile, Italy, and Japan. Firstly, the model was trained using a combination of these datasets. Then, fine-tuning was carried out on each of the target datasets. The model underwent a two-step training process, enabling it to extract overarching features from the merged datasets and subsequently adjust to the unique attributes of each individual target dataset. Another deep CNN model has been developed for magnitude estimation for EEWS proposed by [237]. That DCNN model relied on some EQ properties such as frequency, amplitude, and energy. The model considered 3 s following the P-wave onset from a dataset from Japan.

8.4. Phase association

Phase association is among the prominent issues facing EQ signal processing. It concerns the sequential events triggers reaching observatories and not be easy to decide whether they refer to the same event, a different event, or even not an event. Diffidently, such a global problem affects the EQ catalogs. Indeed, the problem with phase association is clear when we deal with a dense seismic network such as the one relying on IoT nodes.

PhaseLink is an innovative framework in seismic phase association that utilizes state-of-the-art advancements in deep learning [238]. PhaseLink stands out by connecting phases that originate from the same seismic event, rather than using typical grid-based methods. The framework's primary advantage resides in its training procedure, in which it acquires the ability to link phases by analyzing millions of artificially manufactured sequences of P and S wave arrival timings, created based on a 1-D velocity model. The effectiveness of PhaseLink is proven by its exceptional performance on seismic sequences from southern California and synthetic sequences from Japan. These sequences are purposely created to evaluate the framework's performance under extreme conditions, demonstrating PhaseLink's accuracy in linking seismic phases even when the origin times of the events are only about 12 s apart. Additionally, a technique called Gaussian Mixture Model Association (GaMMA) has been suggested [239]. GaMMA integrates a Gaussian mixture model with EQ location, origin time, and magnitude estimate. It treats EQ phase association as an unsupervised clustering problem inside a probabilistic framework. Every EQ is regarded as a grouping of P and S phases, distinguished by a hyperbolic change in arrival times and a decrease in magnitude as distance increases. The multivariate Gaussian distribution represents the set of phase selections for an event, where the mean is the projected arrival time and amplitude resulting from the causal event.

The task of associating seismic phases turns out to be a more intricate job than initially anticipated. It is crucial to embrace and integrate uncertainty in order to improve the process of phase association. Moreover, it is essential to use exceptionally potent and sophisticated deep learning models to address the intricacies inherent in the travel time patterns of seismic waves [240]. Furthermore, utilizing supplementary physical data alongside arrival times emerges as a potential approach to improve the efficiency of deep learning models in addressing the intricacies of seismic phase connection.

8.5. Source discrimination

A key use is categorizing seismic occurrences, which includes EQs, explosions, and other types of ground motion. Supervised learning techniques enable the classification of these occurrences using labeled datasets, enabling the differentiation between natural EQs and generated seismicity or other disturbances. Feature extraction is essential in training ML models to accurately distinguish between different data points. It involves capturing important characteristics such as amplitude, frequency content, and duration [40]. Unsupervised learning approaches, alongside supervised learning, play a role in identifying patterns in seismic data. Clustering techniques facilitate the detection of intrinsic patterns and similarities within seismic datasets, leading to a more profound comprehension of the underlying structures without the need for pre-established labels. ML-based anomaly detection is utilized to identify seismic events or patterns that differ from the predicted norm, providing significant insights into unexpected occurrences [241].

ML techniques have proven to be advantageous in seismic imaging, especially in the field of seismic tomography [242]. Seismic tomography utilizes techniques, such as neural networks, to rebuild underground structures using seismic wave velocities. Moreover, ML assists in distinguishing between various materials that have varied seismic velocities, hence improving the precision and intricacy of subsurface imaging [243]. The application of ML in seismology is crucial for automating seismic monitoring [244]. When ML models are implemented for real-time analysis, they allow for the automatic identification and categorization of seismic occurrences. This feature is essential for the advancement of EEWS, offering prompt notifications to vulnerable regions and perhaps reducing the impact of seismic disasters.

The incorporation of seismic data with additional geophysical and environmental sources enhances the ability to differentiate and distinguish between different elements. ML models demonstrate exceptional

Table 10
Clustering algorithms comparison (Ob = # objects, Cl = # clusters, It = # iterations, DD = Density and distance, H = Hierarchy, K = Kernel, and G = Graph).

	DBSCAN	HDBSAN	K-means	Agglomerative	AP
Clustering based	DD	DD	K	H	G
Advantages	- Suitable for data of different types	- Suitable for data of different types - Suitable for high dimensional feature space	- Suitable for data of different types - Suitable for high dimensional feature space - Able to assess noise and detect overlapping clusters	- Suitable for data of different types - The hierarchical connection between clusters is a quietly easy to get.	- Obvious and Simple - Invulnerable to very high change -Prior to that, # clusters are undesirable
Limitations	- Unequal density of data space and low quality- Clustering strongly variables dependent	-Unequal density of data space and low quality - Not noise-resistant - Clustering strongly variables dependent -Impacts of chaining have an effect	-Not suitable for big deal of data -Clustering is strongly variables dependent -High complexity initially	-An extensive complexity -Planning is fundamentally desired in terms of # clusters.	-An extensive complexity - Clustering strongly variables dependent
Complexities	$O(Ob^{*}logOb)$	$O\left(Ob^{2}\right)$	$O(It^{*}Cl^{*}Ob)$	$O\left(Ob^{3}\right)$	$O\left(Ob^{2^{*}}logOb\right)$

performance in the process of integrating data from several sensors, hence enhancing the precision and dependability in distinguishing seismic occurrences [245]. ML applications in seismology face challenges such as maintaining data accuracy, understanding model reasoning, and adapting to changing seismic circumstances. This highlights the importance of flexible and continuously improving models.

ML techniques have been suggested as a means of differentiating between EQs and QBs [162,204,26]. Significantly, this study confirmed the efficacy of using the catalogs to effectively carry out the discrimination task. Convolutional and recurrent neural networks have been proposed as viable methods for differentiating between local EQs and QBs [246]. The suggested model was trained using a 5-year event catalog from the University of Utah Seismograph Stations. It was trained to automatically distinguish between these two categories using 90-s event spectrograms obtained from both three-component and single-channel seismograms. Furthermore, the CNN network has been employed to effectively differentiate between EQs and explosions by utilizing the generalizability of physics-based properties. This method consists of two branches: one that uses physics-based parametric parameters and the other that uses deep learning to analyze seismic waveforms or spectrograms directly. The characteristics included high-frequency P/S amplitude ratios and the difference between the coda duration magnitude and local magnitude. The outstanding capacity to adapt the model to new locations with high accuracy results from the combination of these elements.

8.6. Clustering techniques

Conventional approaches to grouping seismic events frequently encounter difficulties in managing the large quantity and wide range of seismic signals, hence impeding the proper categorization of EQs. ML algorithms, such as unsupervised learning approaches like k-means, hierarchical clustering, and Density-Based Spatial Clustering of Applications with Noise (DBSCAN), are extremely useful for automatically detecting patterns and groups in seismic datasets.

K-means clustering sorts seismic data into categories defined by common waveform features, magnitudes, and locations. In comparison, hierarchical clustering provides a more complex view of seismic relationships by organizing seismic events into clusters in a hierarchical structure. The density-based method DBSCAN is able to manage the heterogeneous features of seismic activity, making it an excellent tool for detecting seismic clusters of varying sizes and shapes. EQ catalog is just one aspect of ML-enhanced clustering in seismology. In addition, fault structures are defined, aftershock sequences are identified, and aberrant seismic activity is diagnosed. ML algorithms have an adaptive quality that enables them to consistently acquire knowledge and adjust to changing seismic patterns, hence enhancing the precision of cluster allocations as time progresses. Both empirical and theoretical approaches can be utilized to calculate the local site response. Several research endeavors offer explanations of various methods for assessing the specific effects on a local site, based on the topographical and geological characteristics of the investigated area [247,248].

Currently, seismologists are increasingly employing ML methodologies in their research. The potential encompasses the discovery of concealed patterns and signals, as well as the identification of attributes that could enhance our comprehension of geology. A comprehensive examination of ML methods is provided in [249]. In various domains of computer science assessment, the identification of categories within data is an essential initial stage. These issues can be resolved using algorithms [250]. Clustering is an unsupervised data mining strategy that tackles the task of dividing a specific set of items into meaningful subgroups [251].

Table 10 provides a separate classification of clustering algorithms influenced by AI. This table outlines the advantages and disadvantages of various clustering algorithms commonly used in seismology.

8.7. Intensity prediction

Seismic hazard maps (SHMs) illustrate the spatial distribution of dangers resulting from an EQ, specifically the ground motion. Thus, measurements such as PGA and peak ground velocity (PGV) are commonly employed in conventional seismic hazard assessments as indicators of potential risk. On a global scale, a significant quantity of SHMs have been manufactured and proposed. Planners, engineers, and developers commonly utilize SHM systems to reduce expenses and prevent loss of life [252]. Furthermore, they are employed in the fields of DM, seismic risk assessment, and construction rules. Liquefaction, landslides, and tsunamis are potential consequences that may occur following an EQ. Consequently, the potential sites of these effects can be identified by integrating SHMs with supplementary information obtained from tectonic maps, geological, geodetic, and geophysical datasets [41]. The soil classification in Daejeon, South Korea, was mapped using GIS interpolation techniques to create a system of SHMs. These SHMs were developed utilizing RS and geotechnical data. The incorporation of ML into the field of seismic engineering has become a potent and inventive method for forecasting and examining crucial ground motion characteristics, including PGA, PGV, and peak ground displacement (PGD). Historically, seismic evaluations predominantly depended on empirical models and deterministic methodologies. However, the emergence of ML techniques has introduced a new age characterized by improved accuracy and efficiency. ML algorithms can analyze large datasets of recorded seismic events and their corresponding ground motion characteristics to identify complex patterns. This analysis helps in making more accurate forecasts of parameters such as PGA, PGV, and PGD.

PGV, PGA, and PGD are crucial metrics for EQ catastrophe management, seismic hazard analysis, and risk assessment. The researchers in [253] evaluated the accuracy of PGA forecasts for dynamic rupture

Table 11
ML role for EQ parameters observation for EWS.

Ref	Aspect	Application	Description
[264]	Event Classification	Supervised Learning	Categorizing seismic events (EQs, explosions) based on labeled datasets using predefined classes.
[265]	Feature Extraction	Supervised Learning	Capturing seismic characteristics (amplitude, frequency, duration) for training models in event discrimination.
[266,267,268]	Unsupervised Learning	Clustering	Identifying inherent patterns and similarities in seismic datasets without predefined labels.
[268,269,270]	Anomaly Detection	Anomaly Detection	Recognizing seismic events or patterns that deviate from the expected norm, aiding in identifying unusual occurrences.
[271,272,273]	Magnitude Estimation	Regression	Predicting EQ magnitudes based on observed features, facilitating quantitative assessment of potential impact.
[18,181,182,183,274]	Location and depth estimation	Regression	Predicting EQ epicenter or hypocenter
[275,276,277,200]	PGA, PGV, PPV, and intensity	Regression and Classification	Enhancing accurate decision taken by EWS
[278,279]	Seismic Imaging	Seismic Tomography	Reconstructing subsurface structures using algorithms, such as neural networks, to discriminate between different materials with varying seismic velocities.
[280,281]	Automation of Seismic Monitoring	Real-time Analysis	Deploying ML models for automated detection and classification of seismic events, supporting the development of EEWS.
[282,283]	Multisensor Data Fusion	Integration with Geophysical Data	Combining seismic data with other geophysical and environmental sources to improve discrimination accuracy through multisensor data fusion.
[284,285,286]	Challenges	Data Quality, Interpretability, Evolving Seismic Conditions	Ensuring data quality, interpreting model decisions, and addressing the dynamic nature of seismic conditions over time.

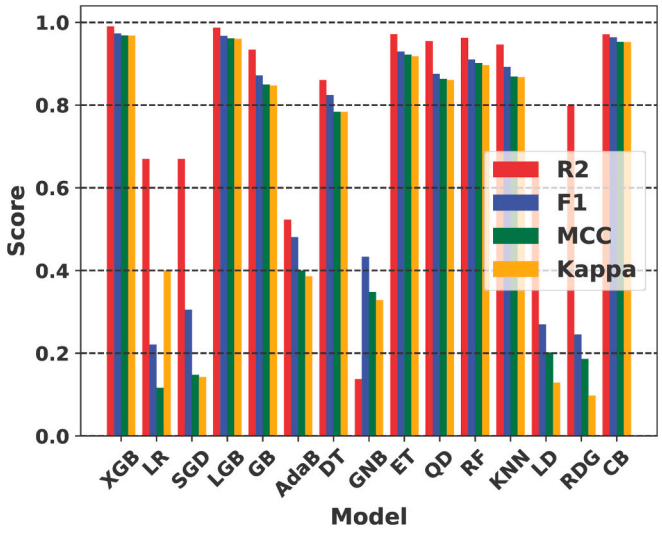


Fig. 9. Comparison of accuracy of on-site intensity classification.

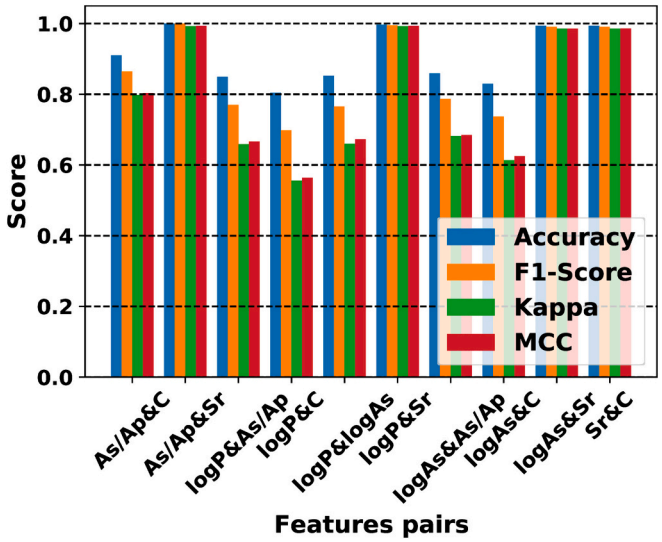


Fig. 10. Comparison of accuracy of EQ and QB classification based on feature pairs.

situations by employing two locking models and various ground-movement models, in addition to real events. The model considered the PGV and PGD measurements of M7 EQs. The degree of stress variability is influenced by the orientation and dimension of the breach. In a study conducted in Chile, recurrent neural networks were employed along with a combination of ground motion intensity to provide a

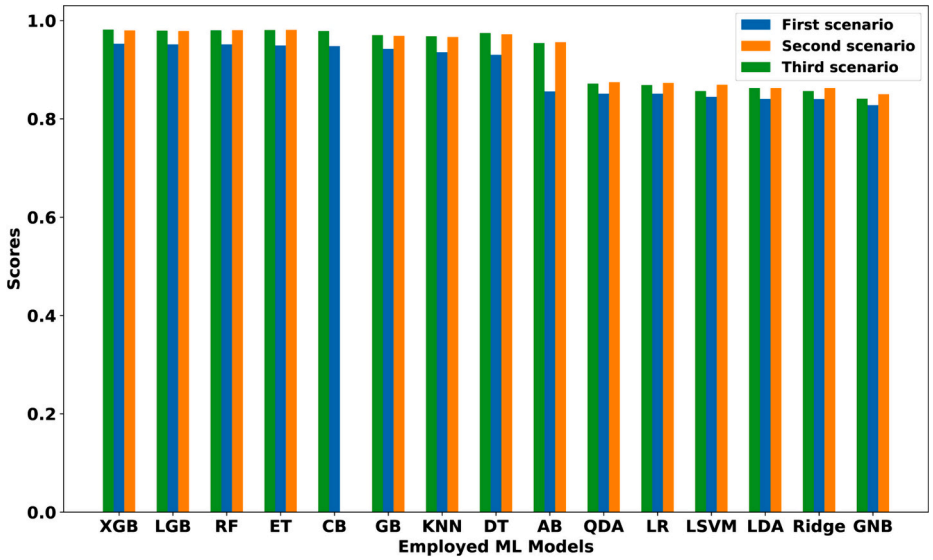


Fig. 11. Linear and Nonlinear ML models accuracy comparison using three scenarios for main shock, aftershock, and QB classification.

comprehensive representation of the scene. This approach was described in reference [254]. This model utilized more than 7000 ground motion records obtained from 1700 occurrences. These findings are beneficial for choosing ground motions that are compatible with dangers for the nonlinear time history analysis of a complex finite-element model. Multiple endeavors have been undertaken to examine the hazards associated with EQ.

According to [255], the investigations have demonstrated the use of ground motion algorithms in calculating the expected levels of seismic ground shaking. These algorithms can be applied for magnitudes ranging from M2 to M7.7. Moreover, as analyzed in [256], precise monitoring of fault rupture data can be employed to compute the PGA and severity level. The study considered a total of 1048 EQs in California, Oregon, and Washington, with magnitudes ranging from M2.7 to M7.1. PGV monitoring has also contributed to the assessment of seismic threats. A hybrid deep learning model, as described in reference [257], has been proposed for the purpose of determining the on-site PGV. Furthermore, the functionality of DL has been broadened to include the estimation of ground acceleration at the location. The authors of [258,259] utilized specific ground motion data to calculate the acceleration response spectrum. The model has undergone three previous EQs to assess its effectiveness, and the findings indicate an 85 % accuracy rate in predicting EWS. The main seismic events and subsequent smaller tremors, measured using spectral accelerations, are important factors in the precise evaluation of intensity. The authors of [260] employed an LSTM model to accurately determine the intensity of the PGA, as well as identify the mainshocks and aftershocks, achieving an impressive accuracy rate of around 85 %.

8.8. Earthquake early warning systems

The incorporation of ML techniques with EEWS has become a pioneering frontier in the advancement of EQ preparedness and response measures. ML algorithms, powered by extensive datasets of seismic activity, have an unparalleled chance to transform the speed and precision of EQ predictions. The integration of ML with EEWS has the capacity to not only improve the accuracy and promptness of early warnings but also represent a fundamental change in our capability to reduce the effects of seismic occurrences on communities and infrastructure [16,15,261].

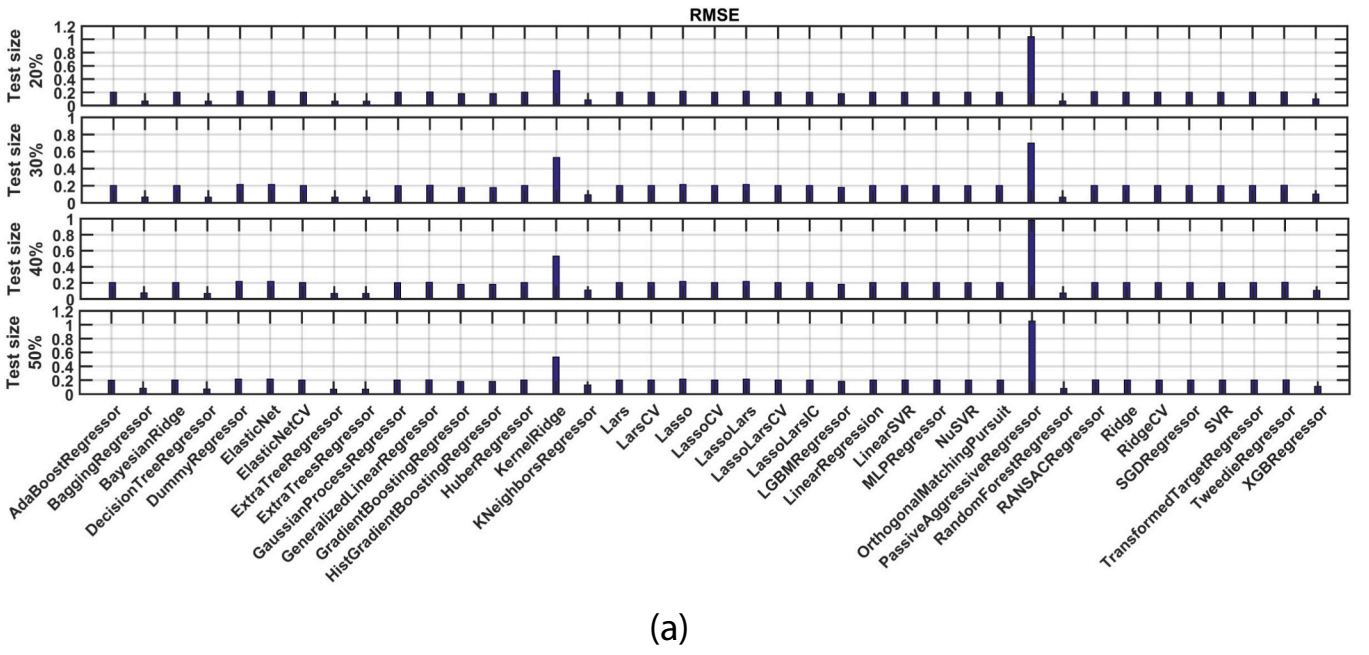
The convergence of AI and seismology signifies a significant advancement in the pursuit of building societies that are more equipped to withstand and respond to the unpredictable nature of EQs. The utilization of seismic phase type and transmission direction has been employed to address the problems of delayed alerting and incorrect triggering of events outside of the intended network. This is essential for the effective deployment of an appropriate EWS. The authors in [262] used a tiny array to determine two properties of local events with ML 2.0–4.6: the back azimuth and the phase delay. They employed three techniques known as beam packing, frequency-wave number, and plane-wave fitting.

A post-seismic EWS has been investigated in a previous study [263]. To predict five future events in the Los Angeles region, the proposed model relies on data regarding the size, location, and magnitude of EQs ranging from M3 to M7.3 that have occurred in southern California. The obtained data indicated that a nonergodic site reaction may be utilized to alert larger areas. Again, Fig. 7 depicts an integrated EWS that utilizes ML to examine and analyze data collected by contemporary data technologies. The monitoring process commences with the utilization of RS, facilitated by observatories. Next, the analysis of the observed data commences by taking into account all computations of EQ parameters.

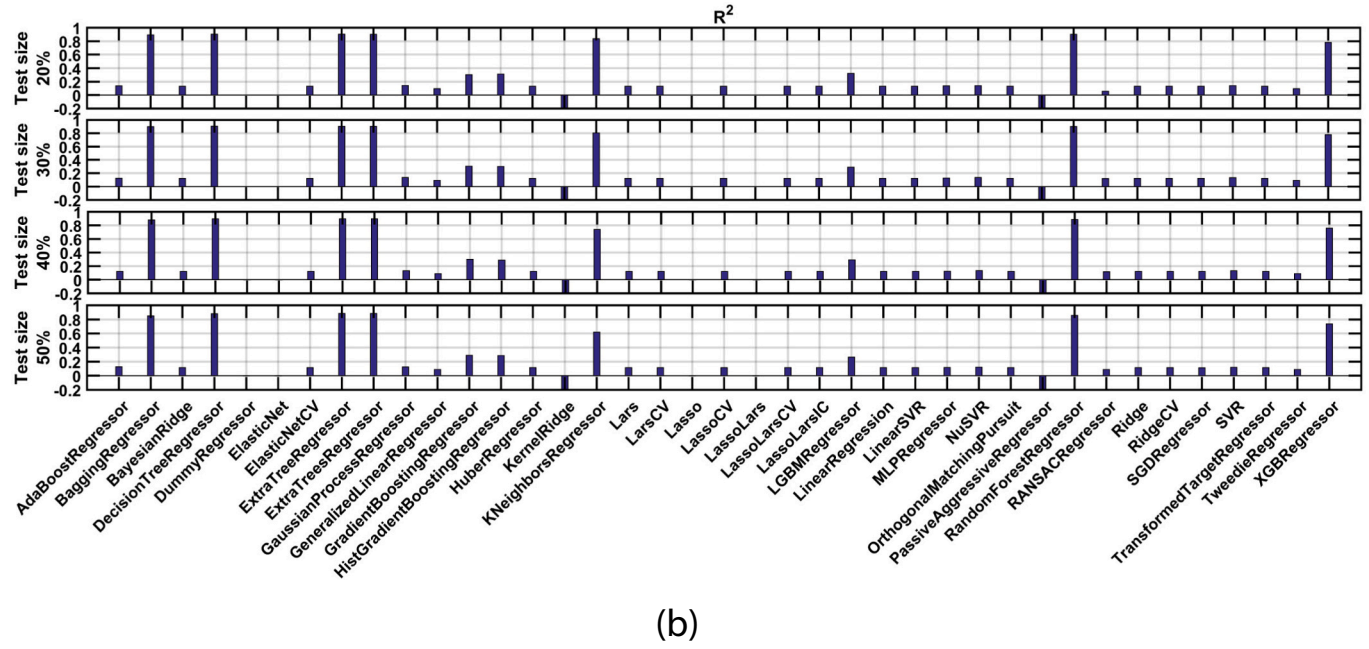
Table 11 displays a thorough classification of the primary parameters of EQs that are necessary for a dependable and efficient EWS. The Seismic Assessment of Monitoring System (SAMS) depicted in Fig. 7 is responsible for detecting, selecting, and associating seismic phases, among other tasks. The purpose of this system is to provide the necessary information to interconnected organizations in the EWS for the evaluation of seismic hazards, as well as for pre-disaster, during-disaster, and post-DM, as mentioned in references [18,42]. This system leverages the dependability and swift responses of the IoT network in conjunction with the cloud system for accurate detection, effortless setup, adaptability, little energy usage, and reliability, among other benefits. Specifically, smart meters can have a substantial impact in this context [287].

8.9. Landslide

The relationship between landslides, EQs, and EWSs is critical for mitigating natural disasters and protecting lives and infrastructure. EQs often act as a triggering mechanism for landslides, particularly in regions with steep terrain and unstable geological conditions. The sudden shaking and ground movement during an EQ can dislodge rocks, soil, and debris, leading to landslides that can cause significant damage and pose severe risks to communities. EWSs play a vital role in this context by providing timely alerts of imminent seismic activity, which can enable individuals and authorities to take precautionary measures.



(a)



(b)

Fig. 12. Performance evaluation of the test set (a) RMSE and (b) R2 using best features combination $log\left(\frac{d}{\sqrt[3]{Q}}\right)$ concerning the test sizes (20 %, 30 %, 40 %, and 50 %).

Modern technologies significantly enhance the effectiveness of these EWSs. Advanced seismographs and accelerometers detect seismic waves almost instantly, providing critical lead time before the shaking reaches populated areas. Satellite monitoring and remote sensing technologies offer real-time data on ground deformation and changes in the landscape that may indicate a heightened risk of landslides. Ground-based sensors and geotechnical instruments measure soil moisture, slope stability, and other factors that influence landslide susceptibility. Additionally, sophisticated computer models and algorithms analyze vast amounts of data to predict the potential impact and spread of landslides following an EQ. These technologies, combined with efficient communication networks, ensure that accurate and timely warnings reach emergency responders and the public, facilitating prompt evacuations and other risk-reduction measures. By integrating modern technologies into EWSs, we can greatly enhance our ability to anticipate and mitigate the effects of these interconnected natural hazards.

More particularly, the use of ML in the study of landslides and the development of solutions to prevent them has provided new opportunities to comprehend and tackle the intricate mechanisms of slope instability [164]. Landslides are frequently influenced by a variety of elements, such as geological, hydrological, and meteorological circumstances. Conventional approaches, although useful, may confront difficulties in comprehending the complexities and non-linear connections inherent in such a complicated issue. ML methods, such as decision trees, random forests (RF), and CNN, are highly proficient in analyzing and acquiring knowledge from extensive datasets [220]. This makes them ideal for landslip prediction and monitoring and for identifying trends and correlations linked to landslip occurrence. For this purpose, the algorithms can analyze a wide range of geographical and environmental factors, such as soil type, rainfall distribution, land surface, and

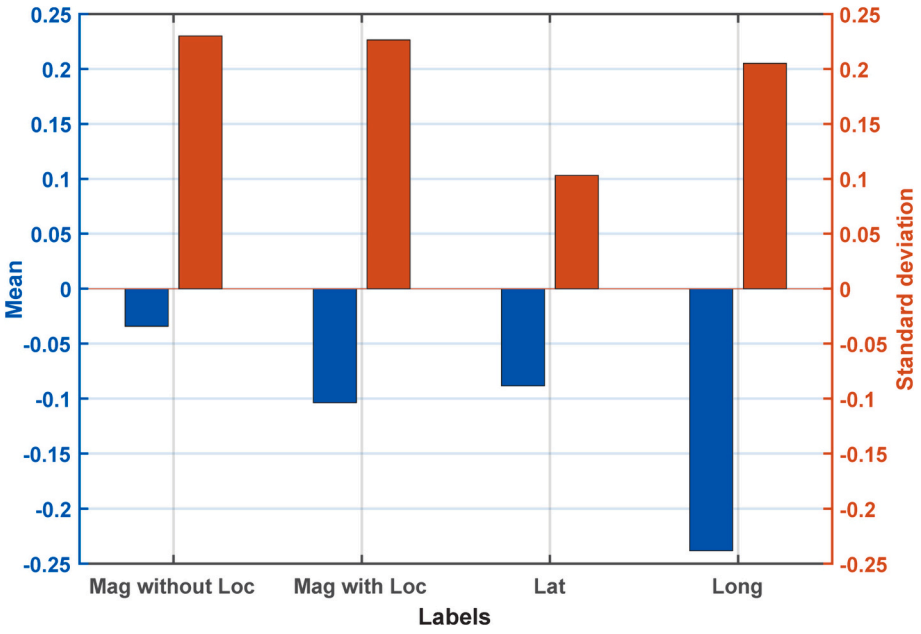


Fig. 13. Comparison of μ and σ of the estimated latitude, longitude, and magnitude.

terrain. In addition, these algorithms learn continuously from both historical landslip occurrences and real-time observation, which allows them to provide more accurate forecasts and important insights into the changing stability of slopes.

Furthermore, ML can be used to create EEWS in landslip scenarios. Indeed, these technologies can alert communities and authorities in advance by seeing early indicators in the data, enabling them to take preventative measures to lessen the impact on people and infrastructure. We can now better understand, predict, and mitigate the dangers associated with slope instability because of the integration of ML with landslip research. Improved landslide resistance and more sustainable management of landslide-prone areas may be within our reach as these technologies develop further. Lastly, in order to preserve human life and infrastructure, landslides must be considered in the EWS.

8.10. Limitations of AI for seismology

Consider the limitations of AI when using it for seismic data analysis and EQ study, despite its immense potential in several domains, including seismology. Some important caveats of AI as it pertains to seismology are these [288,289,290,291,8]:

1. Data availability and quality: AI algorithms often require large amounts of high-quality data for training and validation. In seismology, obtaining a comprehensive dataset with diverse seismic events, including rare or extreme events, can be challenging. Additionally, the quality and consistency of seismic data can vary due to factors such as sensor limitations, noise, and data gaps. These limitations can impact the performance and reliability of AI models, especially when dealing with uncommon or poorly represented seismic events.
2. Domain expertise requirement: Seismology concepts and domain knowledge are frequently necessary for accurate seismic data interpretation and analysis. Even though AI systems can figure out how things work by analyzing data, they cannot have the same level of context or expertise as a seismologist. However, it still needs more research effort.
3. Limitations in generalization: sometimes AI has limited capabilities to get novel solutions when training complex dataset. For example, when AI deals with different seismological settings such as variable tectonic regimes, different locations or geographical areas only a single data set is not enough to get sufficient results. Accordingly, building a model to fit such a problem needs to be adaptable to be generalized.
4. Lack of trustworthiness and interpretation: It is probable to get results that do not make sense or convey a logical manner especially when dealing with deep learning models. In other words, the obtained results have uncertainty, especially, when the model is designed for forecasting without a thorough understanding of seismic signals. Accordingly, this point is still under investigation.
5. Large resources for computation: In the case of multidimensional problems in seismology, high-processing capabilities are needed to establish an effective model that can deal with a variety of data sources such as images, time series, etc. Accordingly, large-scale seismic datasets will need complicated AI algorithms.
6. Ethical and social issues: some AI models may have a lack of interpretation of the actual reason for false alarm of EWS due to attacks or human misbehavior. Such a problem may need human intervention. Accordingly, it can reduce the advantage of automation.

Overall, ML and DL have made significant strides in seismology by improving many different tasks and applications. Seismic wave denoising, seismic area clustering, accurate EQ parameter determination, peak particle velocity prediction, acceleration of seismic waves detected, seismic event classification, estimation of incomplete seismic data, and EEWS development have all been made possible by the utilization of AI techniques. These applications highlight the vast potential of AI in seismology, enabling stakeholders to make more informed decisions and mitigate the impact of EQs on human life and infrastructure.

9. AI for applied seismology using real datasets

In this section, we explore two main directions of applying AI techniques to seismological data analysis - classification and regression. For each direction, we present three case studies that utilize different real-world seismic datasets. Through these case studies, we demonstrate the capabilities of AI methods in tasks such as EQ detection, magnitude estimation, and other seismic signal processing applications. We also provide comparative figures to highlight the performance and tradeoffs of the proposed AI approaches.

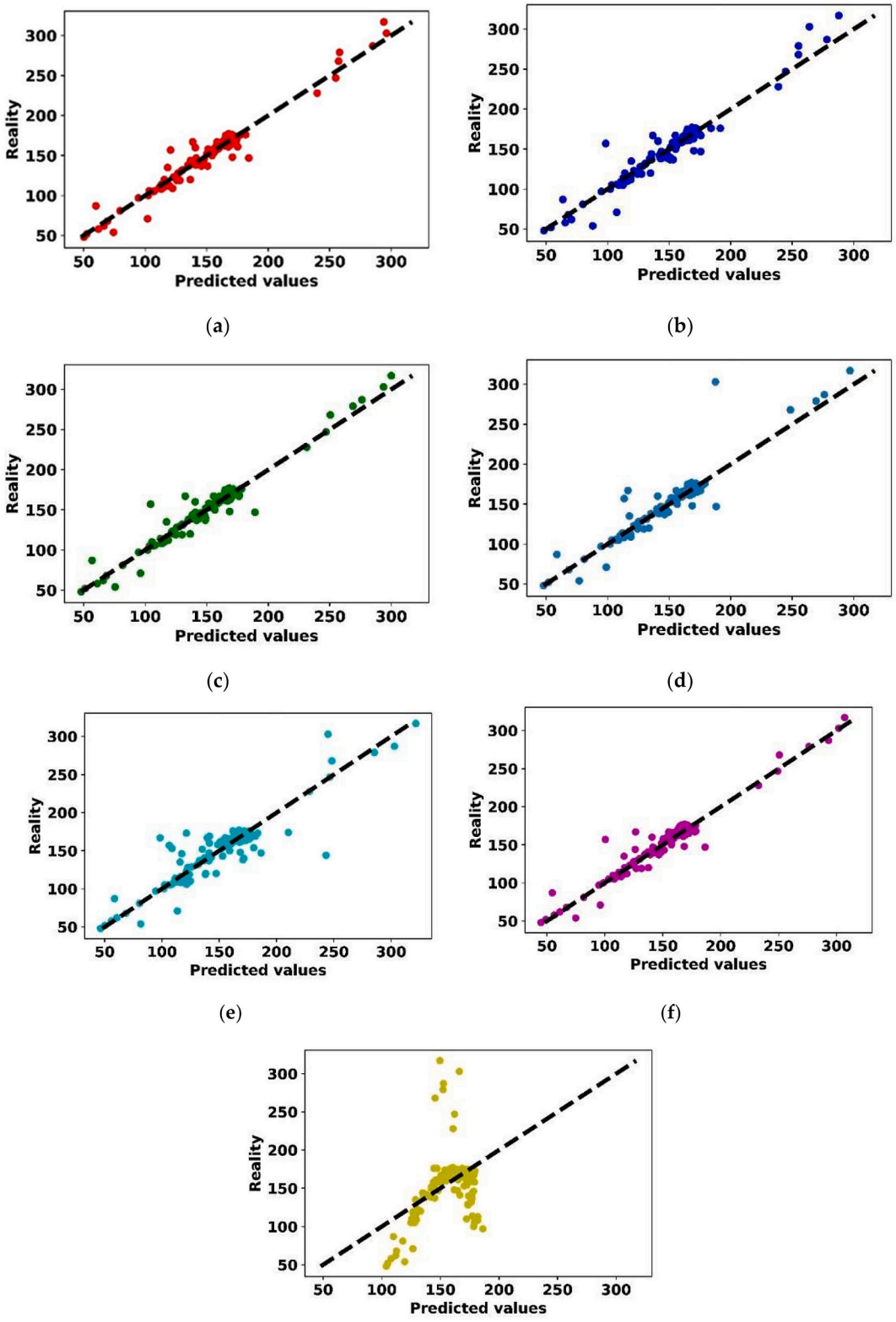


Fig. 14. Q-Q curve for the estimated PGA using (a) ML and other benchmarks (b) IDW method, (c) Kriging, (d) Natural method, (e) Spline method, (f) TopoToR method, and (g) Trend method.

Table 12
Modern technologies advantages and limitations for EQ disaster.

Technology	Type of Data Used	Advantages	Limitations
Remote Sensing	Optical imagery, radar data (Synthetic Aperture Radar - SAR), LiDAR data, thermal images, and multispectral/ hyperspectral data [292,293,294].	High-resolution data, real-time monitoring, and the ability to cover inaccessible areas	Limited temporal resolution and dependence on weather conditions (e.g., cloud cover can obstruct optical sensors).
Autonomous Robots	Visual data (cameras), LIDAR data, infrared sensors, sonar, GPS data, and environmental sensors [295,296,297].	Ability to operate in dangerous or difficult-to-access areas, reducing human risk, and high precision in data collection.	Limited by battery life, terrain navigation challenges, and initial deployment costs.
UAVs	High-resolution visual images, video footage, thermal imagery, LiDAR data, and multispectral data.	Quick response time, high mobility, and ability to provide detailed imagery and video from various angles.	Limited flight duration, payload capacity, and weather dependency.
Signal Processing	Seismic waveforms, acoustic signals, accelerometer data, and other geophysical data [298].	Enhanced detection capabilities, ability to filter out noise, and improved accuracy in identifying seismic activity.	Requires sophisticated algorithms and high-quality data input.
AI	Historical seismic data, real-time sensor data, satellite imagery, and geological data [299,300,301].	High processing power, ability to identify patterns and anomalies, and continuous learning to improve predictions.	Requires substantial data for training, potential for algorithmic errors, and high computational resources.
Social Media	Text posts, images, videos, geotagged information, and user-generated reports [302,303].	Wide reach, rapid information dissemination, and crowdsourced data collection.	Information can be unreliable or unverified, and there can be a high noise-to-signal ratio.
IoT	Sensor data (vibration, pressure), GPS data, and communication logs [304,305,306].	Real-time data collection, integration with other systems, and automated response capabilities.	Dependence on connectivity, potential cybersecurity risks, and maintenance of numerous devices.

9.1. AI for seismic parameter classification

In this subsection, we investigate the application of AI techniques to the classification of seismic signals. We present three case studies that leverage ML algorithms to tackle seismic signal classification tasks using diverse real-world datasets.

1. **Seismic Intensity Estimation for EEWS** [275]**:** The main contribution of this work is the proposal of an EEWS based on an ML model, which quickly determines EQ intensity using limited features of seismic waveform traces. The study presents the outcomes of hyper-parameter tuning, demonstrating the key findings that resulted in the best model with the most accurate categorization of six intensities. Fig. 9 illustrates the performance metrics, where the "XGB" model achieves the best results with 98.59 % R2, 96.7 % F1, and 96.03 % and 96.02 % of both Kappa and MCC, respectively. The dataset used in this study is collected from the Italian national seismic network, consisting of 386 stations and 50,000 occurrences with 150 thousand 2 s-three-component seismic windows.
2. **Augment Analyst Decisions for Seismic Source Discrimination** [26]**:** The main contribution of this paper is a classification scheme that minimizes the number of features while achieving 100 % successful discrimination between EQs and QBs. The model relies on only two features, leading to a linear function output similar to logistic regression. Fig. 10 shows the comparison of various ML models using accuracy, F1-score, Kappa, and MCC. The best model is achieved by the "XGB" that employs the tuple of two features, $\{As/Ap, Sr\}$, with 100 % accuracy and F1 score, and 99.3 % Kappa and MCC. The second-best model is the "RF" with the tuple $\{logP, Sr\}$, achieving 99.6 % accuracy, 99.5 % F1-Score, and 99.3 % Kappa and MCC.
3. **Performance Assessments for Seismic Sources Discrimination** [162]**:** The main contribution of this research is the development of a novel discrimination scheme using ML models to accurately differentiate between artificial seismic sources and EQs. Fig. 11 compares the obtained accuracy of three scenarios: three classes (main shocks, aftershocks, and artificial seismic sources), two classes with de-clustering (main shocks and artificial seismic sources), and two classes without de-clustering (EQs and artificial seismic sources). The best performance is obtained using the third scenario, with 98.14 % accuracy achieved by the "ExtremeGradientBoosting" model, while the best linear model "QuadraticDiscriminantAnalysis" attains 87.17 % accuracy. The second-best performance is obtained using the second scenario with 98.1 % accuracy by the "ExtraTreesClassifier" model.

9.2. AI for seismic parameter regression

This subsection focuses on the use of AI methods for the regression of seismic parameters. We present three case studies that demonstrate the potential of AI-driven regression techniques in extracting valuable insights from complex seismic datasets.

1. **A Regression Model to Predict Blast-Driven Ground Vibrations** [200]**:** The main contribution of this work is the development of an optimized ML regression model to predict blast-driven ground vibrations, specifically the PPV values in a limestone QB area. The dataset used in this research was compiled from a limestone QB located southeast of Helwan City, Egypt, and includes data from 1438 QB incidents with parameters such as the maximum charge per delay, the distance from the QB location to the monitoring point, and the PPV. The authors compare the obtained results of the RMSE using four test sizes of 39 regression algorithms in Fig. 12 (a), noting that better results can be obtained using a test size of 30 %. Similarly, Fig. 12 (b) portrays the obtained results of R^2 relying on the input data scenario of $log\left(\frac{d}{\sqrt[3]{Q}}\right)$ considering the four test sizes. It is worth mentioning that the obtained results in Figs. 12 (a) and 12 (b) depend on the default training parameters of the examined 39 regression algorithms without optimizing the hyperparameters.
2. **Earthquake Parameters Observation in IoT System-based EEWS** [18]**:** The main contribution of this work is the development of an IoT system-based EEW using a DL model that integrates AE and CNN to swiftly calculate earthquake parameters like magnitude and location. The dataset used in this study consists of 4900 EQ records collected from western Canada over 3 years. The proposed approach aims to achieve minimal error between the predicted EQ parameters and manual solutions by expert seismologists. Fig. 13 shows both the standard deviation (σ) and the mean (μ) for the magnitude-only case and the magnitude with location case. For the magnitude-only case, σ is < 0.19, and for the magnitude with location case, σ is $_c$ for magnitude, < 0.1 for latitude, and < 0.2 for longitude. Furthermore,

the mean error in the predicted magnitude increases from 0.02 to 0.09 when the location is considered along with the magnitude, and the mean error in latitude and longitude is 0.089 and 0.22, respectively.

3. **Identifying Exposure of Urban Area to Certain Seismic Hazard** [37]**:** The main contribution of this research is the development of an adaptive and reliable model using ML and GIS interpolation methods to predict the spatial distribution of PGA caused by the 1992 Cairo EQ in the Greater Cairo area. The dataset used in this study includes Landsat images from 1992, geotechnical data, PGA data from 2005, and population data. The authors employed both the MSE and R^2 values for a valuable visual presentation of the predicted PGA values and compared them with the true values represented by the linear curve, as shown in Fig. 14. The proposed ET ML model achieved the best-predicted PGA values with minimal residuals compared to six other classical methodologies, as illustrated in Fig. 14.

10. Conclusion and recommendations

EQs are among the most devastating natural catastrophes, requiring integrated, intelligent, and reliable solutions. Despite the numerous research efforts in the literature, addressing this phenomenon still demands a prominent solution that can mitigate risks to human life, effectively handle day-to-day issues, and protect vulnerable areas from EQ disasters. Therefore, the efficient utilization of rapidly advancing technologies and techniques becomes crucial for EQ. DM. This article highlights the important roles that signal processing techniques, sensors, RS, robots, UAVs, social networks, IoT, optimization, and AI can play in various aspects of seismology, to alleviate the EQ disaster.

Firstly, the importance of incorporating modern technologies into seismology is emphasized, as traditional methodologies in the field have inherent vulnerabilities that can be addressed through the integration of advanced technologies. The software and hardware components of these modern techniques are described in detail. More particularly, the role of emerging networks in seismology is discussed, highlighting the significance of each element. Thirdly, the importance of signal processing for waves has been addressed. Fourthly, the modernization of sensors has been extensively discussed. Fifthly, various aspects related to RS, autonomous robots, and UAVs are addressed concerning the transmission of real-time monitoring data and transmission reliability. Sixthly, the benefits of utilizing fast booming networks in seismology are underscored, including the utilization of IoT and social media platforms to enhance data collection and dissemination. Seventhly, the correlation between optimization techniques and seismology is explored. Two categories of optimization are discussed: optimization techniques for analyzing seismic signals and optimization methods for extending the lifespan of seismic networks. Lastly, a comprehensive overview of the applications of AI in seismology is presented, showcasing the wide range of areas where AI can be employed in the field.

In light of the current system's vulnerability and the need for further improvements and research collaboration, several recommendations are put forth for future directions. These recommendations encompass the utilization of modern technologies, such as ML, in order to detect EQ. Additionally, the rapid growth of the IoT can be leveraged to address vulnerabilities in the conventional monitoring system. Furthermore, the role of EEWS can be enhanced. Lastly, an integrated system that combines ML, IoT, and optimization techniques can be implemented to facilitate effective DM and risk mitigation. There are three distinct stages to the installation of this system. The first stage relies on an EEW-supported basic alarm, the second stage includes DM, and the third stages are devoted to risk reduction.

Our recommendation for stakeholders is to give more attention to the use of AI for EQ detection using more advanced algorithms that can deal with the complexity of the earth's structure. Moreover, integrating both weak motion ave. and string motion one can support more understanding fro the EQ phenomenon and its prediction. We also recommend merging the GPS, and magnetic data recordings with the seismic data to portray a more concise picture of the earth's structure, enhance the DM, and alleviate the EQ consequence.

Moreover, broadening the use of the dense seismic network via IoT nodes would facilitate the monitoring process as a real-time application. The node can contribute to a better understanding of structural response and environmental variables. Accordingly, it can support seismologists with precise recognition of the seismological settings of different environments.

The power of social networks is a double-edged weapon. So, it should be efficiently utilized to extend the consciousness of people to draw a smooth plan that can be used during the EQ disaster. Moreover, the stakeholder can use it to update their database regularly, especially for the afflicted areas by EQ disaster. Accordingly, the decision-maker can pinpoint the more disastrous area as the higher priority to serve. In this regard, the different communication systems can play a significant role in alleviating the EQ disaster.

Finally, IoT, ML, UAV, social networks, and optimization techniques can work under the same umbrella of a generic system to enhance DM and risk mitigation. More particularly, achieving real-time EQ detection needs well-trained ML algorithms by different data types collected from several sources, which can support the tiny IoT nodes with a simple function to do in a stage of the EEWS. Besides, the UAV can support this system in the far field with real-time monitoring. Accordingly, this integration can reveal a robust and reliable EEWS that can facilitate the decision-maker's mission. To sum up, Table 12 illustrates the strong and weak points of utilizing modern technologies for EQ natural disaster mitigation.

CRediT authorship contribution statement

Mohamed S. Abdalzaher: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Moez Krichen:** Writing – review & editing, Writing – original draft, Validation, Supervision, Resources, Methodology, Investigation, Conceptualization. **Francisco Falcone:** Writing – review & editing, Visualization, Supervision, Resources, Investigation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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